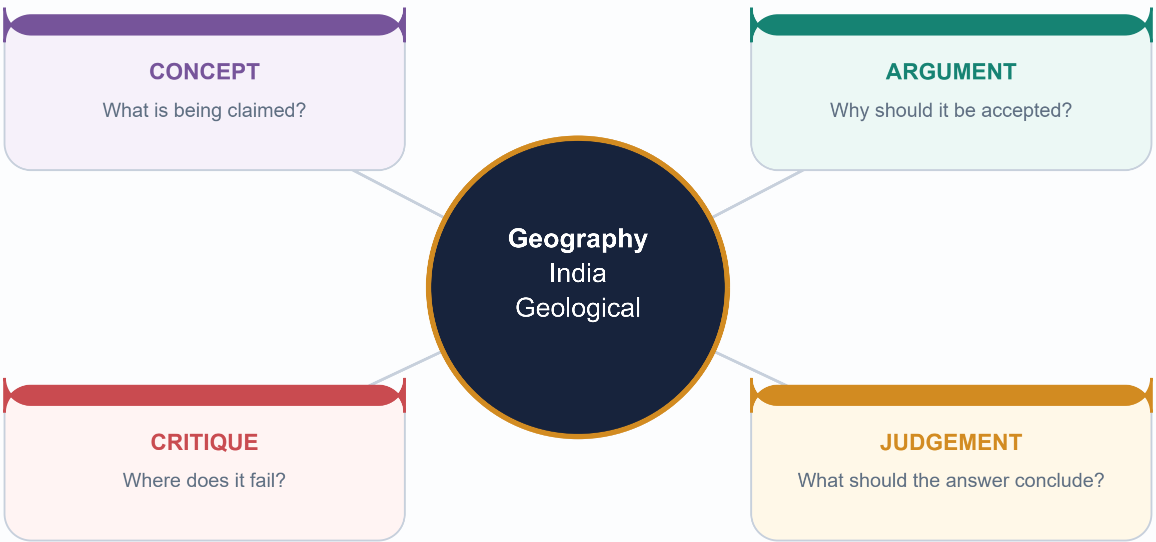
COMPLETE LEARNING SESSION

Geography 02 - India Geological Structure

GS-I + Prelims | Complete Topic Package | 2026-08-15
Repository Markdown -> local OCR books -> bounded official current context
[FACT] verified | [ANALYSIS] synthesis | [LIMIT] caution/ownership
All maps and cross-sections are original, schematic and not to scale.



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

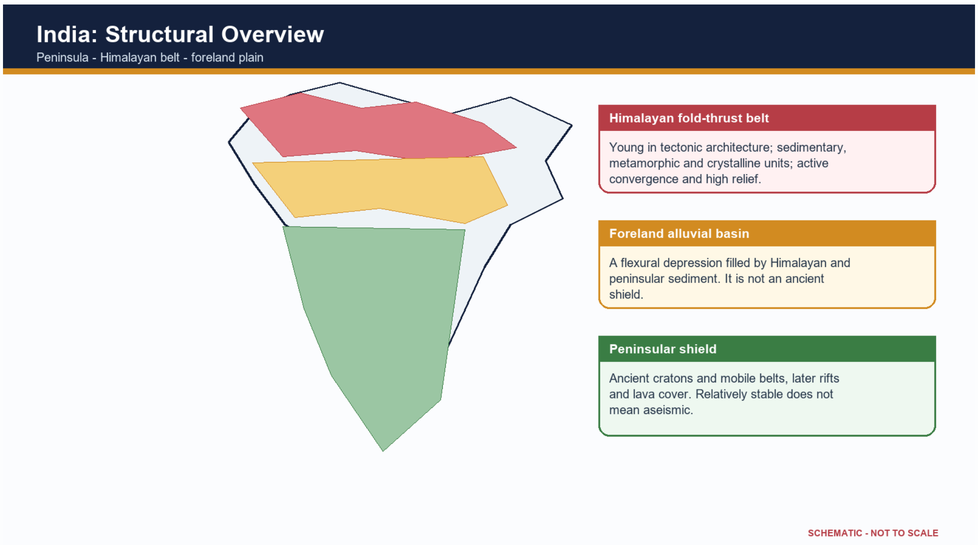
1. Scope: geological structure versus physiography and geomorphology

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] Geological structure means the age, lithology, stratigraphic order, folds, faults, joints, basins, cratons and plate-tectonic relations of rocks beneath a region. Physiography describes the broad surface divisions; geomorphology explains the processes and forms of relief.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Term	What it asks	India example
Geology	rock age, type, structure and tectonic history	Dharwar belts, Gondwana basins, Himalayan thrusts
Physiography	broad relief division	plateau, mountains, plain, coast, islands
Geomorphology	landform process and evolution	escarpment, gorge, waterfall, delta

STATIC THEORY

- [ANALYSIS] The three are linked but not interchangeable. The Peninsular Plateau is a physiographic division; its cratons, mobile belts, rifts, sedimentary basins and basalt cover are geological structure; its scarps, residual hills, waterfalls and valleys are geomorphic expression.

- [FACT] A useful national framework has three interacting provinces: the ancient Peninsular shield, the active Himalayan fold-thrust belt, and the Indo-Ganga-Brahmaputra sediment-filled foreland basin.

- [ANALYSIS] Structure operates as an inherited template. Climate, weathering, rivers, vegetation and human action then modify the template. Therefore geology explains pattern and constraint, not every present landform by itself.

- [LIMIT] Later Geography topics own detailed physiographic divisions, drainage evolution, soils, minerals and disaster management. This package supplies their geological mechanism and clearly identifies those cross-links.

- [ANALYSIS] Exam method: identify the hidden structure, name a region, explain the mechanism, state a surface/resource/hazard consequence, then qualify the relationship.

MUST-KNOW FACTS

1. [FACT] Peninsula = ancient composite shield, not one uniform rock mass.

2. [FACT] Himalaya = young fold-thrust system containing sedimentary, metamorphic and crystalline units.

3. [FACT] Northern plain = alluvial fill in a foreland setting, not exposed ancient basement.

UPSC TRAPS

WRONG: Geological structure is the same as physiography.

CORRECT: Structure is subsurface architecture; physiography is broad surface form.

WRONG: The stable peninsula is structurally inactive everywhere.

CORRECT: It is relatively stable but retains rifts, faults and intraplate seismicity.

MAINS ANGLE

[ANALYSIS] Begin an answer by separating structure, relief and process; this prevents the common error of turning a geology question into a list of physiographic divisions.

STUDY LINK

Geography Topics 03-11 for hazards and landforms; Topic 31 for resources.

HIGH

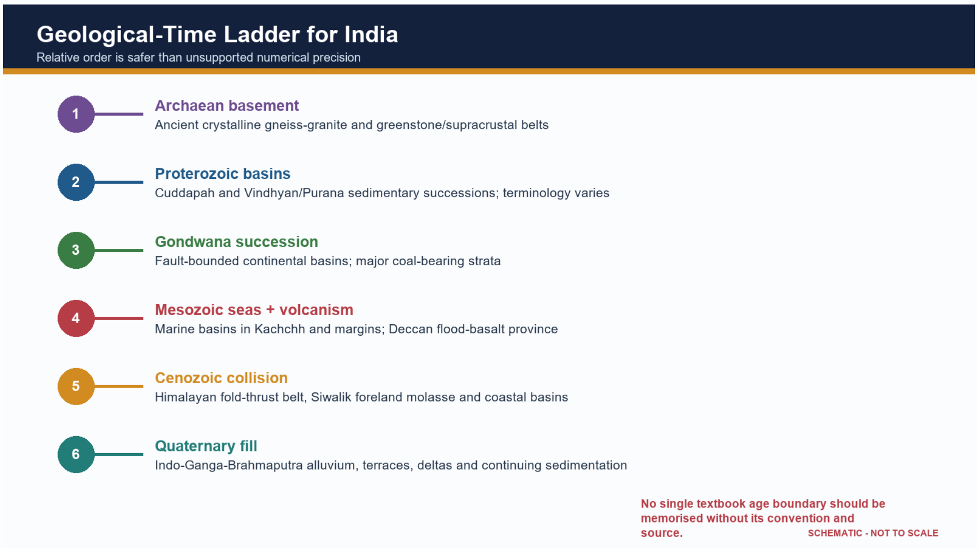
2. Geological time, evidence and dating cautions

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] India's geological record spans ancient crystalline basement, Proterozoic basins, Gondwana continental successions, Mesozoic marine and volcanic episodes, Cenozoic collision-related strata and Quaternary alluvium.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Evidence	What it can establish	Caution
Lithology/structure	rock character and relations	similar rocks can recur at different ages
Fossils	relative age and environment	not available in all units
Radiometric age	numerical event age	may date metamorphism/cooling, not deposition
Palaeomagnetism	past latitude/rotation and polarity	overprinting must be tested
Seismic imaging	subsurface geometry	interpretive, resolution-dependent

STATIC THEORY

- [FACT] Relative dating uses superposition, cross-cutting relations, unconformities, fossils and correlation. It orders events without necessarily assigning a numerical age.

- [FACT] Radiometric dating estimates numerical ages from isotope decay systems in suitable minerals. A date may record crystallisation, metamorphism, cooling or later alteration; it must be interpreted geologically.

- [FACT] Fossils are especially useful in sedimentary correlations, but many ancient Indian crystalline and Proterozoic successions are unfossiliferous or sparsely fossiliferous.

- [FACT] Palaeomagnetism records past field directions and supports plate reconstruction, sea-floor spreading and movement of the Indian plate.

- [ANALYSIS] Geological maps integrate field relations, petrography, geochemistry, geochronology and geophysics. Seismic reflection, gravity and magnetic methods extend interpretation beneath alluvium, sea and lava cover.

- [LIMIT] Textbook systems differ: older Indian schemes use labels such as Purana, Dravidian and Aryan differently from modern international chronostratigraphy. State the scheme before using it.

- [LIMIT] Numerical boundaries and formation ages are revised as dating improves. Relative order is safer in UPSC unless a precise number is source-verified.

MUST-KNOW FACTS

1. [FACT] Unconformity = missing time/erosional or non-depositional gap.

2. [FACT] A radiometric date needs an event interpretation.

3. [FACT] Stratigraphy is not merely a rock-name list.

UPSC TRAPS

WRONG: One isotopic date gives the age of the entire rock system.

CORRECT: The dated mineral and geological event must be specified.

WRONG: Archaean, Precambrian and Purana are always exact synonyms.

CORRECT: Usage varies by author and stratigraphic convention.

MAINS ANGLE

[ANALYSIS] Use an evidence ladder: field relation -> dating/correlation -> geophysical image -> tectonic interpretation -> qualification.

STUDY LINK

Physical Geography Core Topic 02; Science and Technology cross-link for geophysical methods.

HIGH

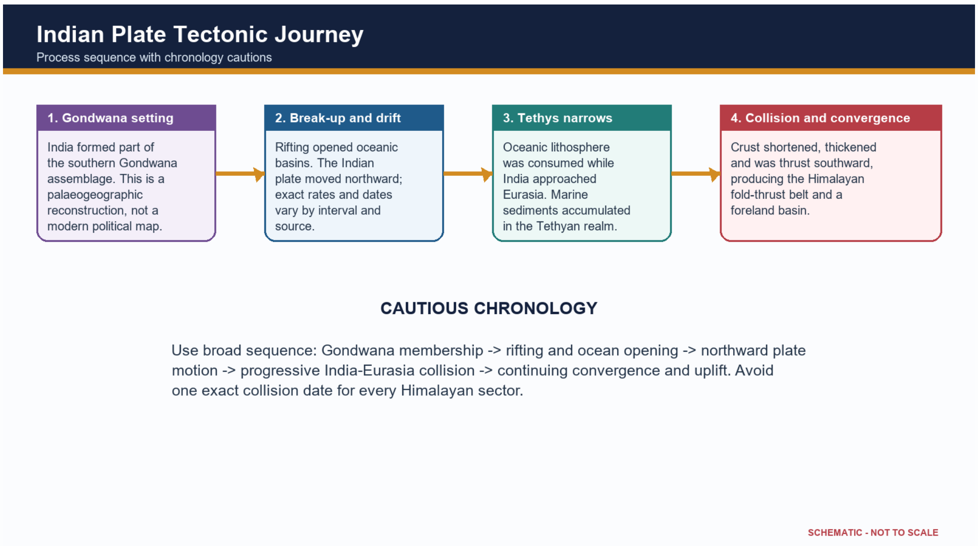
3. Indian plate: Gondwana origin, break-up, drift and collision

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] The Indian continental block formed part of Gondwana. Rifting separated blocks, oceanic crust formed between them, and the Indian plate moved northward before colliding progressively with Eurasia.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Stage	Process	Indian geological result
Gondwana assembly	continental continuity	shared shield and sediment evidence
Rifting	faulting and basin opening	passive-margin and rift-basin development
Northward motion	Tethys narrowing	marine sediment and arc/suture records
Collision	shortening and thickening	Himalayan thrust belts and Tibetan uplift
Post-collision	erosion and flexure	Siwalik molasse and foreland alluvium

STATIC THEORY

- [FACT] Continental fit, matching ancient rock belts, fossil and sediment correlations, palaeoclimatic indicators and palaeomagnetism together support former continental connections.

- [FACT] Sea-floor spreading creates oceanic lithosphere at ridges; subduction consumes it at trenches. The disappearance of Tethyan oceanic domains and India-Eurasia convergence produced crustal shortening and the Himalayan-Tibetan orogen.

- [ANALYSIS] Collision was not a single instant across the whole margin. Different sutures, terranes, sedimentary basins and metamorphic events record a prolonged and spatially variable process.

- [FACT] Continuing convergence is expressed through active thrusting, earthquakes, uplift, erosion and sediment transfer into the foreland and offshore basins.

- [LIMIT] Avoid an unsupported constant plate speed or one exact collision date. Values depend on the time interval, plate circuit and definition of initial versus hard collision.

- [ANALYSIS] India is both an ancient continental fragment and part of an active plate system: this explains the contrast between old shield rocks and young Himalayan tectonic architecture.

MUST-KNOW FACTS

1. [FACT] Gondwana is a palaeocontinent; Gondwana Supergroup is an Indian stratigraphic succession.

2. [FACT] Himalayan collision thickens crust rather than creating a volcanic arc like the Andes.

3. [FACT] Tethys was an oceanic realm, not one unchanged narrow sea.

UPSC TRAPS

WRONG: Gondwanaland and Gondwana coal measures are identical terms.

CORRECT: One is a palaeocontinent; the other is a named sedimentary succession.

WRONG: The Himalaya is a subduction volcanic chain.

CORRECT: It is primarily a continent-continent collision fold-thrust belt.

MAINS ANGLE

[ANALYSIS] Present chronology as a process chain and explicitly distinguish evidence from reconstruction.

STUDY LINK

Core Topic 02 crustal dynamics; Topic 03 earthquakes; Topic 06 Himalayan glaciers.

HIGH

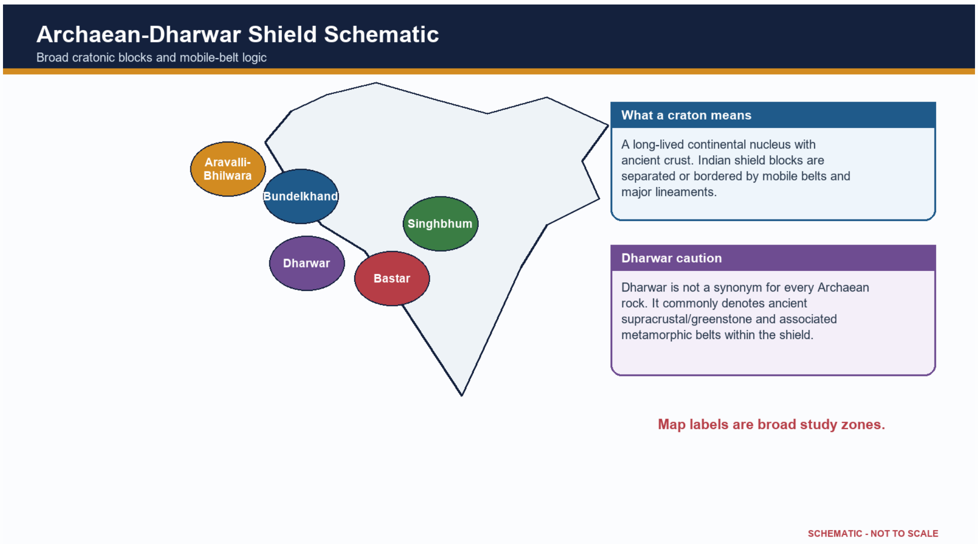
4. Cratons, shields and the stabilised Peninsular block

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] Peninsular India is a mosaic of ancient cratonic nuclei and intervening mobile belts, later cut by faults, rifts and dykes and locally covered by younger sediments and basalt.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Block/belt	Broad region	Exam-safe cue
Dharwar	Karnataka and adjoining south	greenstone/metamorphic belts and metallic-mineral associations
Bastar	Chhattisgarh-central India	ancient shield with mineral and basin margins
Singhbhum	Jharkhand-Odisha	iron-ore and shear-zone associations
Bundelkhand	UP-MP	granite-gneiss basement and residual uplands
Aravalli-Bhilwara	Rajasthan-west	ancient deformed belt and non-ferrous mineral provinces

STATIC THEORY

- [FACT] Commonly recognised shield blocks include Dharwar in the south, Bastar in central India, Singhbhum in the east, Bundelkhand in north-central India and Aravalli-Bhilwara in the west. Boundaries and nomenclature differ among geological maps.

- [FACT] Cratons preserve old continental crust; mobile belts record deformation, metamorphism and accretion between or around them.

- [ANALYSIS] Long denudation exposes basement rocks and concentrates resistant residual uplands. Yet later tectonics produced the Narmada-Son, Cambay, Godavari and other structural corridors.

- [FACT] Ancient crystalline rocks commonly have low primary porosity; groundwater occurs mainly in weathered mantles, joints, fractures, faults and locally permeable contacts.

- [ANALYSIS] Metallic-mineral provinces are strongly associated with ancient shield geology, but ore localisation requires specific magmatic, hydrothermal, sedimentary or metamorphic processes.

- [LIMIT] Stable means lower average deformation than an active plate boundary over recent geological time. It does not mean flat, homogeneous, immobile or earthquake-free.

MUST-KNOW FACTS

1. [FACT] Shield = exposed ancient continental crust; craton can include covered stable crust.

2. [FACT] Peninsular basement is composite.

3. [FACT] Secondary permeability is vital in hard-rock aquifers.

UPSC TRAPS

WRONG: All peninsular rocks are Archaean.

CORRECT: Proterozoic basins, Gondwana strata, Deccan basalt and younger sediments also occur.

WRONG: Mineral wealth follows age alone.

CORRECT: Ore-forming processes and later preservation/localisation are necessary.

MAINS ANGLE

[ANALYSIS] Use 'mosaic' rather than 'single block'; then connect old crust to relative stability, minerals, hard-rock groundwater and residual relief.

STUDY LINK

Topic 31 mineral distribution; Topic 04 groundwater; Environment mining impacts.

HIGH

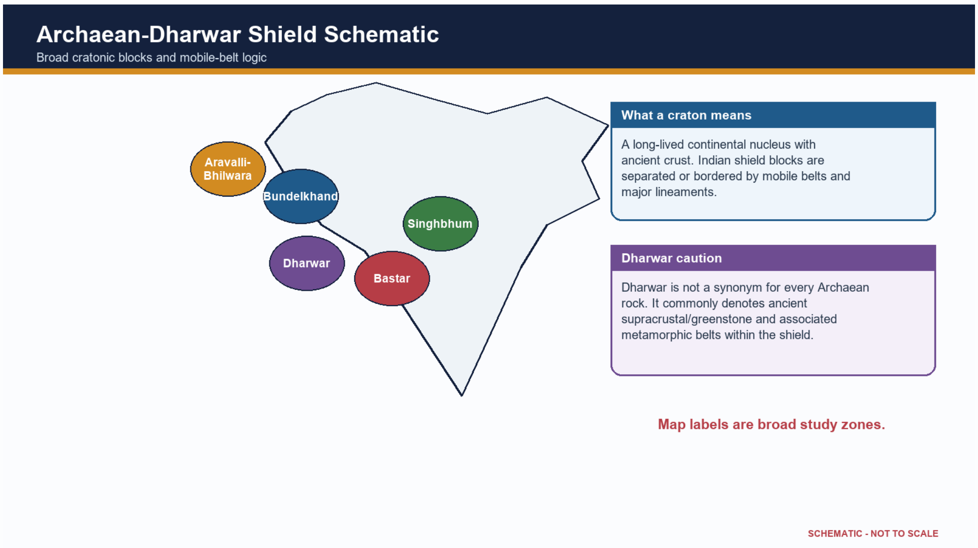
5. Archaean and Dharwar systems: distribution and terminology control

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] Older Indian geography texts distinguish fundamental gneiss-granite basement from Dharwar supracrustal and metamorphosed sediment-volcanic belts. Modern geological usage is more refined, so the terms must not be collapsed.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Unit	Exam-safe description	Distribution hook
Fundamental gneiss/granitoid	ancient crystalline basement	wide Peninsular exposure
Dharwar/greenstone belts	metamorphosed supracrustal volcanic-sedimentary belts	Karnataka and other shield provinces
High-grade complexes	gneiss, granulite, charnockite associations	southern and eastern shield sectors
Ore belts	localised mineralised zones	Kolar, Singhbhum, Karnataka-Odisha-central belts

STATIC THEORY

- [FACT] Archaean basement rocks include ancient gneisses, granitoids and high-grade metamorphic complexes across large parts of the Peninsula and in crystalline Himalayan cores.

- [FACT] Dharwar-type belts are best developed in Karnataka and adjoining southern India, with comparable ancient supracrustal/metamorphic belts in Singhbhum, central India, Aravalli-Delhi sectors and the eastern shield.

- [FACT] These belts contain schists, quartzites, iron formations, metavolcanics and associated intrusions; exact stratigraphic grouping varies by geological province.

- [ANALYSIS] Their economic importance derives from repeated magmatism, sedimentation, metamorphism and hydrothermal activity, not merely antiquity.

- [FACT] Iron, manganese, gold and several base-metal associations occur in ancient shield belts. Named examples such as Kolar or Singhbhum are location anchors, not universal rules for every Dharwar rock.

- [LIMIT] Do not repeat old claims that all Archaean rocks are entirely unfossiliferous or that one textbook's age span is definitive. Modern Precambrian geology uses finer subdivisions and evidence.

MUST-KNOW FACTS

1. [FACT] Dharwar is not all Archaean.

2. [FACT] Ancient shield belts are commonly metamorphosed and structurally complex.

3. [FACT] Named ore belts are examples, not deterministic pairings.

UPSC TRAPS

WRONG: Dharwar is a younger Phanerozoic sedimentary system.

CORRECT: It belongs to India's ancient Precambrian shield record.

WRONG: Every gneiss belongs to one formation and age.

CORRECT: Gneiss is a rock/texture term and can form in different events.

MAINS ANGLE

[ANALYSIS] In a classification answer, give lithology, structure, distribution, significance and a terminology caution.

STUDY LINK

Advanced Topic 31; local OCR: Khullar Ch.2 and Majid Geography of India Ch.1.

HIGH

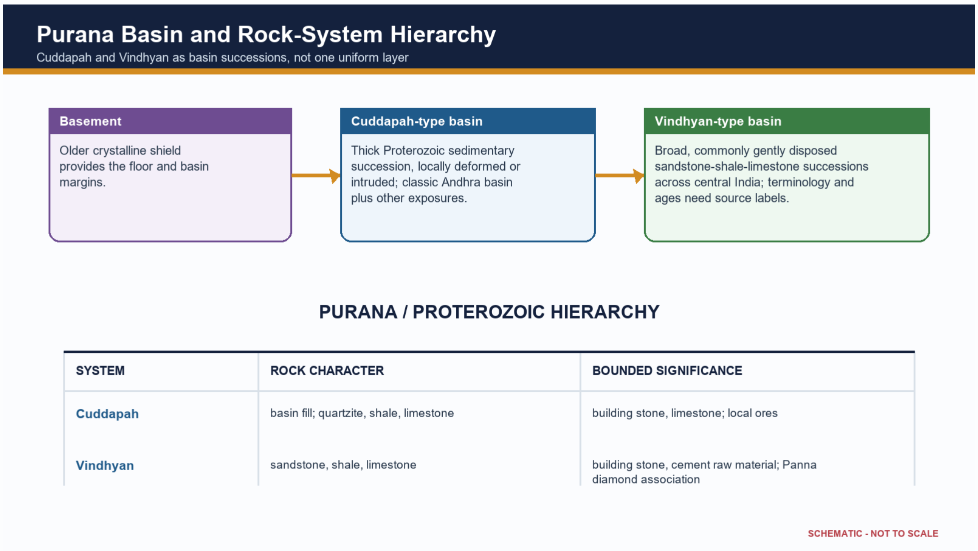
6. Proterozoic Purana basins: Cuddapah and Vindhyan

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] Cuddapah and Vindhyan are major Proterozoic sedimentary basin successions resting unconformably on older basement. Older textbooks group them under Purana, but their detailed ages and correlations are convention-sensitive.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

System	Broad lithology	Distribution	Significance
Cuddapah	quartzite, shale, limestone, sandstone; local intrusions	Andhra basin plus central/western exposures	limestone, stone and local mineral occurrences
Vindhyan	sandstone, shale, limestone	Rajasthan-MP-Son belt and related basins	building stone, cement raw material, Panna association

STATIC THEORY

- [FACT] The Cuddapah basin is classically developed in Andhra Pradesh, with related Proterozoic successions in central and western India. Thick sandstone, shale, limestone and quartzite record long-lived basin subsidence, sedimentation, deformation and intrusion.

- [FACT] Vindhyan rocks form an extensive central Indian succession from eastern Rajasthan through Madhya Pradesh toward the Son valley, with other basin occurrences. Sandstone, shale and limestone dominate.

- [ANALYSIS] Basin thickness and repeated unconformities show episodic subsidence and tectonic reorganisation rather than one uninterrupted quiet deposit.

- [FACT] Vindhyan sandstones and limestones are important building and industrial materials; the Panna diamond association is a high-yield location cue.

- [LIMIT] Resource links must be bounded: a whole system is not uniformly mineralised, and Panna diamonds do not make every Vindhyan sandstone diamond-bearing.

- [LIMIT] Avoid treating Cuddapah and Vindhyan as identical, coeval everywhere or simply 'young sedimentary rocks'. They are old Proterozoic basin records with different internal histories.

MUST-KNOW FACTS

1. [FACT] Cuddapah and Vindhyan overlie older basement.
2. [FACT] They are sedimentary basin successions, not cratons.
3. [FACT] Terminology and age ranges vary across sources.

UPSC TRAPS

WRONG: Vindhyan is a volcanic province.

CORRECT: It is dominantly a sedimentary succession.

WRONG: Cuddapah equals the modern Kadapa district only.

CORRECT: The name originates there but the geological system is broader.

MAINS ANGLE

[ANALYSIS] Explain them as long-lived intracratonic basin archives; do not present an unqualified memorised age table.

STUDY LINK

Topic 08 karst/limestone; Topic 31 resources; History/Art cross-link for building stone only.

HIGH

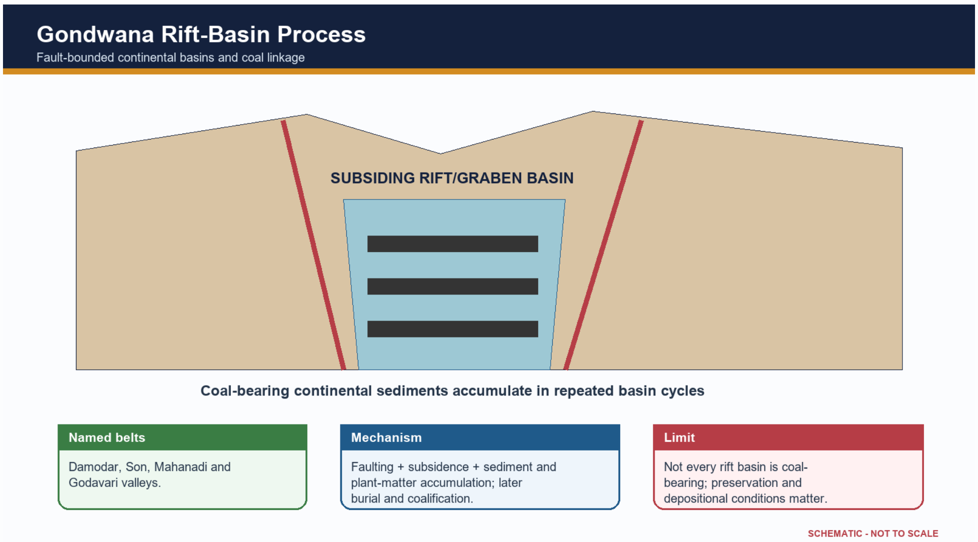
7. Gondwana Supergroup: rift basins, coal and continental sedimentation

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] India's Gondwana successions accumulated mainly in fault-bounded continental basins within the Peninsular shield. They are distinct from Gondwana the palaeocontinent.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Basin belt	Named anchors	Structural cue
Damodar	Raniganj, Jharia, Bokaro, Karanpura	faulted east-central troughs
Son-Mahanadi	Singrauli, Korba, Talcher-Ib	linked intracratonic basins
Godavari	Singareni/Godavari Valley	elongate rift/graben setting
Satpura	Pench-Kanhan and adjoining fields	central structural basin

STATIC THEORY

- [FACT] Major basins follow the Damodar, Son-Mahanadi, Satpura and Godavari structural belts. Faulting and subsidence created accommodation space for river, lake, floodplain and swamp deposits.

- [FACT] Lower parts of the succession include glacially influenced deposits in some basins, followed by coal-bearing sandstone-shale cycles and younger continental strata.

- [ANALYSIS] Coal linkage is mechanistic: vegetation accumulation in suitable low-oxygen settings, burial during basin subsidence, compaction and coalification, followed by preservation from erosion.

- [FACT] Jharia, Raniganj, Bokaro, Talcher-Ib, Singrauli and Singareni are high-yield basin/coalfield anchors. Exact reserve percentages and current production shares belong to dated official resource sources.

- [LIMIT] Gondwana coalfields are not simply surface strips along modern rivers. They occupy structural basins whose present drainage may only partly follow inherited troughs.

- [LIMIT] Coal presence is not automatic in every Gondwana unit or every rift. Depositional environment, burial history and preservation control workable seams.

MUST-KNOW FACTS

1. [FACT] Gondwana coal basins are rift/fault-linked.

2. [FACT] They are continental sedimentary basins.

3. [FACT] Coal association is strong but not universal.

UPSC TRAPS

WRONG: Gondwana coal occurs in the Deccan Trap.

CORRECT: Coal measures are sedimentary and generally predate/underlie or lie outside basalt cover.

WRONG: Gondwana means Palaeozoic only.

CORRECT: The Indian succession extends across a broader late Palaeozoic-Mesozoic interval depending on subdivision.

MAINS ANGLE

[ANALYSIS] A top answer draws a graben cross-section, locates three basin belts and qualifies geology-resource determinism.

STUDY LINK

Topic 31 owns coal/resource distribution and policy; Environment owns mining impact.

HIGH

8. Marine sedimentary basins, coasts and offshore petroleum logic

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] India also preserves marine sedimentary records in Kachchh, Rajasthan, Himalayan Tethyan sectors, east-coast basins and younger coastal/offshore basins.

ORIGINAL STUDY VISUAL

Rock-System - Age - Resource Matrix

Associations are probabilistic, not deterministic

System / setting	Broad age-status	Typical rocks	Bounded resource linkage
Archaean-Dharwar shield	Precambrian; schemes vary	gneiss, granite, schist, greenstone	metallic-mineral provinces; not every belt mineralised
Cuddapah-Vindhyan	Proterozoic / Purana	sandstone, shale, limestone, quartzite	stone, limestone; Panna diamond association
Gondwana basins	late Palaeozoic-Mesozoic succession	continental sandstone, shale, coal measures	major coalfields in faulted basins
Deccan Trap	late Cretaceous-earliest Cenozoic transition	stacked basalt flows	soil, groundwater and construction links; local mineralisation
Cenozoic basins	Tertiary-Quaternary	marine/continental sediment and alluvium	petroleum where source-reservoir-seal-trap coexist

SCHEMATIC - NOT TO SCALE

Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Setting	Examples	Geological significance
Rift/onshore basin	Cambay, Assam-Arakan	faulted subsiding basin and traps
Passive/offshore margin	Mumbai Offshore, KG, Cauvery	thick sediment, delta/margin architecture
Mesozoic marine basin	Kachchh, Jaisalmer sectors	marine fossils and transgression record
Tethyan belt	Ladakh-Spiti and Himalayan sectors	former continental margin/ocean record

STATIC THEORY

- [FACT] Marine transgressions left fossiliferous limestone, sandstone, shale and related strata in Kachchh and other Mesozoic basins. Tethyan Himalayan sequences record environments on the northern continental margin and oceanic realm.

- [FACT] Coastal and offshore sedimentary basins include Cambay, Mumbai Offshore, Assam-Arakan, Krishna-Godavari, Cauvery and other margin basins. Names are map anchors; resource status must come from current official basin data.

- [ANALYSIS] Petroleum requires an effective source rock, burial and maturation, migration, porous reservoir, seal and trap. Thick sediment alone is necessary in many settings but never sufficient.

- [ANALYSIS] Passive-margin subsidence, deltaic sediment supply, rifting and faulting can create favourable basin architecture. Offshore occurrence is therefore geological, not simply because an area is under the sea.

- [LIMIT] Do not claim every named basin is equally productive or that all offshore oil is young. Exploration, economics, technology and environmental constraints modify geological potential.

- [LIMIT] Detailed hydrocarbon distribution is owned by Geography Topic 31. This section explains basin formation and petroleum-system logic.

MUST-KNOW FACTS

1. [FACT] Hydrocarbons are basin resources, unlike most shield-hosted metallic ores.

2. [FACT] Source-reservoir-seal-trap is the minimum answer chain.

3. [FACT] Marine strata also occur inland due to past seas.

UPSC TRAPS

WRONG: All sedimentary basins contain petroleum.

CORRECT: A complete petroleum system and preservation are required.

WRONG: Offshore oil is produced by seawater pressure.

CORRECT: It formed from buried organic matter within sedimentary basin systems.

MAINS ANGLE

[ANALYSIS] Explain offshore concentration through basin evolution, not a memorised field list.

STUDY LINK

Geography Topic 31 petroleum; Environment Topic 24 coastal regulation.

HIGH

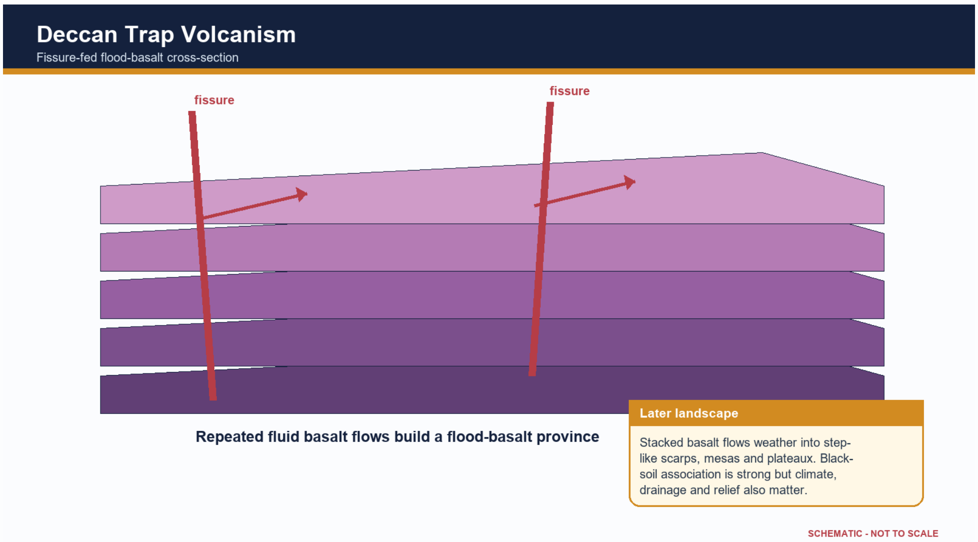
9. Deccan Trap: flood-basalt province and bounded surface linkages

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] The Deccan Trap is a vast flood-basalt province built by repeated, mainly fissure-fed lava flows around the end-Cretaceous to earliest Cenozoic transition.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Feature	Geological cause	Surface/resource effect
Step relief	stacked resistant flows	terraced scarps and plateaux
Interflow zone	weathering/vesicles between flows	local groundwater storage
Dyke/fracture	magma intrusion and later jointing	barrier or conduit depending on condition
Black-soil belt	basaltic parent material plus pedogenesis	moisture-retentive clay-rich soils in many tracts

STATIC THEORY

- [FACT] Stacked basalt flows create step-like topography; 'trap' derives from a word for stair/step. Present exposure is concentrated in Maharashtra and extends into Gujarat, Madhya Pradesh, northern Karnataka and adjoining outliers.

- [FACT] Flow interiors, vesicular tops, weathered horizons, fractures, dykes and intertrappean beds create strong vertical and lateral variation.

- [ANALYSIS] This architecture governs groundwater: compact basalt may transmit little water, while fractures, vesicular zones, weathered contacts and interflow horizons can form aquifers.

- [ANALYSIS] Basalt weathering contributes parent material for black soils, but clay mineralogy, drainage, relief, climate and time control the final soil. Deccan Trap is a volcanic province, not a soil type.

- [FACT] Differential erosion produces plateaux, mesas, buttes, scarps and flat-topped hills. Later river incision and structural lines organise drainage.

- [LIMIT] The cause, duration and environmental effects of Deccan volcanism remain active research fields. Avoid precise eruption pulses or causal claims about mass extinction without a cited scientific source.

- [LIMIT] Natural-resource potential of the Deccan Trap is a direct 2022 GS-I application owned by Topic 31; geology here supplies the mechanism.

MUST-KNOW FACTS

1. [FACT] Deccan Trap = extrusive igneous flood basalt.
2. [FACT] Intertrappean beds occur between flows.
3. [FACT] Black soil linkage is strong but not monocausal.

UPSC TRAPS

WRONG: Deccan Trap is a Gondwana sedimentary system.

CORRECT: It is a younger volcanic province.

WRONG: Basalt is uniformly impermeable.

CORRECT: Aquifer behaviour varies with fractures, weathering, vesicles and flow contacts.

MAINS ANGLE

[ANALYSIS] Structure the answer as volcanism -> flow architecture -> relief/aquifer/soil/resource -> qualification.

STUDY LINK

Topic 04 groundwater; Topic 30 soils/agriculture; Topic 31 Deccan resources.

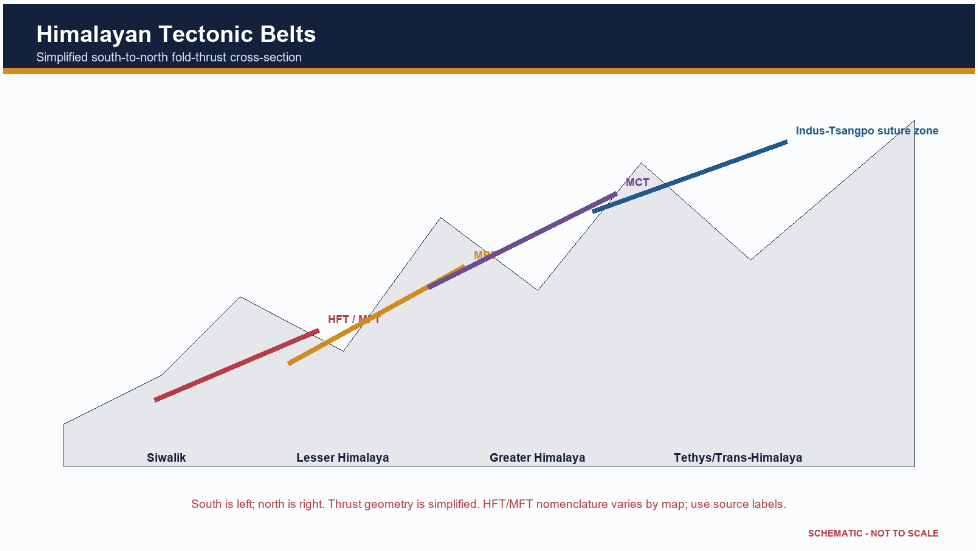
HIGH | 10. Himalaya: from geosynclinal language to modern plate tectonics

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] Older texts describe Tethyan sediment accumulation in a geosyncline followed by folding. Modern plate tectonics explains the driving mechanism through subduction of oceanic domains, collision, shortening, thrusting, metamorphism and uplift.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

South-north belt	Broad character	High-yield boundary
Siwalik/Sub-Himalaya	young foreland molasse, folded foothills	HFT/MFT at front; MBT to north
Lesser Himalaya	older sedimentary/metasedimentary thrust sheets	MBT-MCT zone
Greater Himalaya	high-grade crystalline/metamorphic core	MCT and northern detachments in detailed models
Tethys/Trans-Himalaya	marine sedimentary and magmatic/suture domains	Indus-Tsangpo suture region

STATIC THEORY

- [ANALYSIS] 'Geosyncline' can be retained as a historical descriptive term for a long subsiding sedimentary belt, but not as a self-sufficient causal theory.

- [FACT] The Himalayan orogen includes Trans-Himalayan magmatic/tectonic units, Tethyan sedimentary sequences, Greater Himalayan crystalline rocks, Lesser Himalayan units and the Siwalik/Sub-Himalayan foreland fold-thrust belt.

- [FACT] Major south-verging structures commonly mapped include the Main Central Thrust, Main Boundary Thrust and Himalayan Frontal Thrust/Main Frontal Thrust. The Indus-Tsangpo suture marks a broader collision/suture zone to the north.

- [LIMIT] These structures branch, overlap and vary along strike. A simple north-south classroom cross-section is conceptual and cannot represent every local segment.

- [FACT] Siwalik strata are relatively young foreland molasse derived from erosion of the rising Himalaya and folded at the mountain front. They are not the oldest Himalayan rocks.

- [ANALYSIS] The Himalaya is not one simple fold. It is an imbricated fold-thrust wedge containing nappes, duplexes, faults, metamorphic cores and sedimentary belts.

- [LIMIT] Detailed stratigraphic ages and local thrust order should be quoted only from an appropriate geological map or source; nomenclature can differ.

MUST-KNOW FACTS

1. [FACT] Himalaya = fold-thrust orogen, not one fold.

2. [FACT] Siwalik = foreland molasse belt.

3. [FACT] Greater Himalaya includes crystalline/metamorphic units.

UPSC TRAPS

WRONG: Siwalik is the oldest Himalayan belt.

CORRECT: It is the youngest major outer sedimentary belt.

WRONG: All Himalayan rocks are Tertiary sediments.

CORRECT: Older crystalline, metamorphic and sedimentary rocks are incorporated.

MAINS ANGLE

[ANALYSIS] Contrast the historical geosynclinal description with the modern plate-tectonic mechanism; then draw a labelled cross-section.

STUDY LINK

Topic 03 seismicity; Topic 06 glaciers; Disaster Management landslides.

HIGH

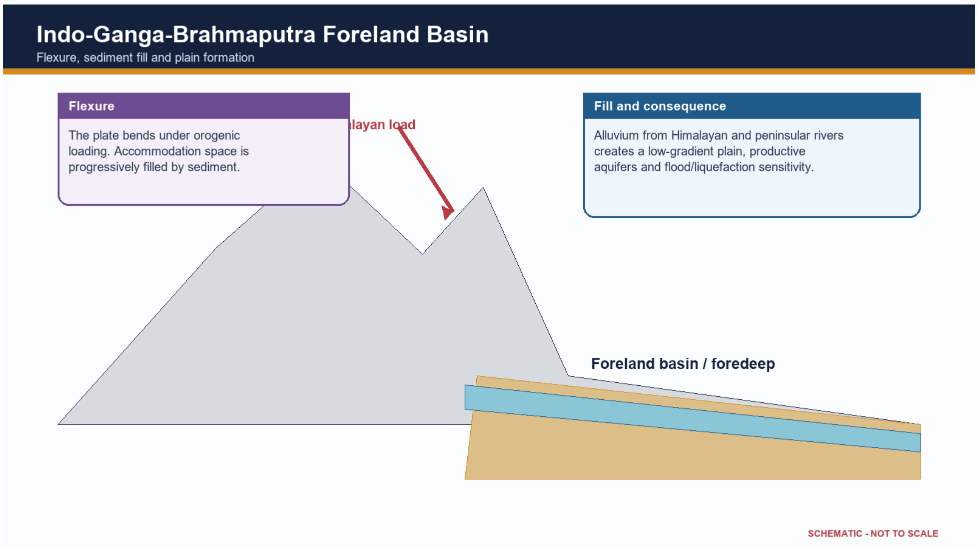
11. Indo-Ganga-Brahmaputra foreland basin and alluvial fill

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] The Northern Plain occupies a foreland depression between the Himalayan orogen and Peninsular basement, filled by large volumes of Cenozoic and Quaternary sediment.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Process	Result	Exam linkage
Flexural subsidence	foredeep/accommodation space	collision-relief connection
Sediment delivery	thick alluvial fill	fertility and aquifers
Channel migration	floodplain and avulsion belts	flood exposure
Soft sediment response	amplification/liquefaction potential	seismic hazard

STATIC THEORY

- [FACT] Orogenic loading flexes the Indian plate downward, creating accommodation space. Rivers from the Himalaya and Peninsula deliver gravel, sand, silt and clay into this depression.

- [FACT] Coarse piedmont deposits grade basinward into finer alluvium. River migration, avulsion, floodplain deposition and subsidence continually reorganise the surface.

- [ANALYSIS] Thick unconsolidated sediment explains level relief, fertile alluvial soils, major aquifers and dense settlement, but also flood, erosion, subsidence and liquefaction sensitivity.

- [FACT] The Siwalik belt represents older foreland molasse now incorporated into the mountain front; the modern alluvial plain lies farther south/east as an actively filled basin.

- [LIMIT] The plain is not one uniform layer or one age. Basement depth, sediment thickness, river history and tectonic influence vary regionally.

- [LIMIT] Detailed bhabar-terai-bhangar-khadar and drainage topics belong later. Here the owner is flexure, sediment fill and structural context.

MUST-KNOW FACTS

1. [FACT] Foreland basin forms by flexure under mountain load.
2. [FACT] Alluvium masks deeper basement.
3. [FACT] Plain stability at surface does not imply low seismic risk.

UPSC TRAPS

WRONG: The Northern Plain is an exposed ancient shield.

CORRECT: It is a sediment-filled foreland basin.

WRONG: Alluvial ground always reduces earthquake damage.

CORRECT: Soft sediment can amplify shaking and liquefy.

MAINS ANGLE

[ANALYSIS] Show the causal chain: Himalayan loading -> flexure -> sediment fill -> plain resources and hazards.

STUDY LINK

Topic 05 drainage; Topic 04 groundwater; Topic 03 earthquake effects.

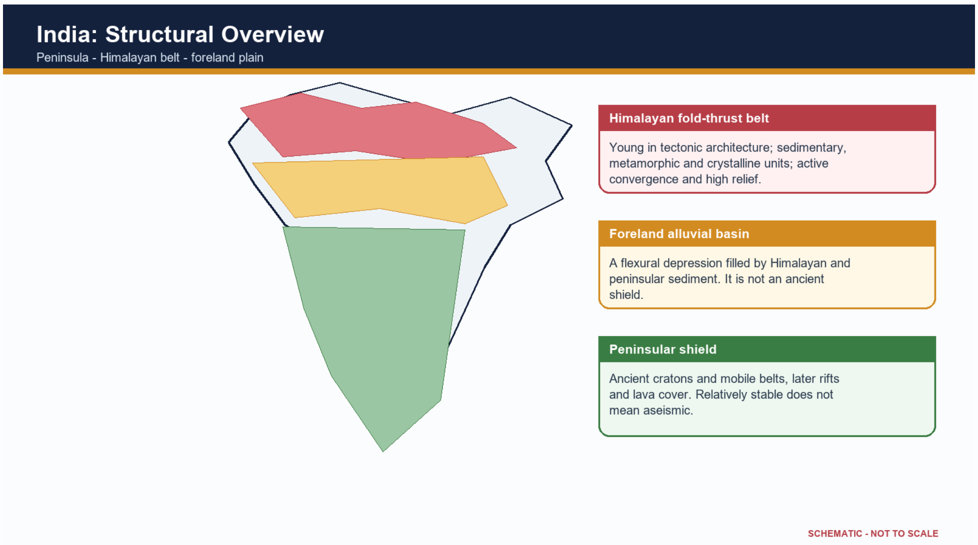
HIGH | 12. Coasts, offshore basins and island-arc versus coral-island distinction

GS-I / Prelims Geography

NEWS TRIGGER

[FACT] India's coasts record rifting, passive-margin subsidence, sedimentation, faulting, volcanism and sea-level change. Its two major island systems have different geological settings.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Island/coast	Foundation	Surface expression	Hazard cue
Andaman-Nicobar	subduction-related arc/forearc	elongate islands, volcano and uplifted/sedimentary units	earthquake-tsunami-volcanism
Lakshadweep	submarine ridge/volcanic foundation	coral atolls and reefs	sea-level, erosion, freshwater lens
West coast	rifted/faulted passive margin sectors	narrow coast, estuaries and basins	subsidence/local seismicity
East coast	sedimentary/deltaic margin basins	broad coastal plains and deltas	cyclone-flood-subsidence cross-link

STATIC THEORY

- [FACT] Western India includes the Cambay rift and faulted margin sectors; parts of the west coast show submergent features. East-coast basins receive large deltaic sediment loads and preserve thick sedimentary successions.

- [FACT] Andaman-Nicobar lies along an active convergent margin/island-arc system linked to the Sunda-Andaman subduction zone and the continuation of the Arakan-Yoma tectonic trend.

- [FACT] Barren Island is an active volcano in this arc setting. The island chain also includes sedimentary and uplifted components; it is not simply a row of volcanic cones.

- [FACT] Lakshadweep consists of coral atolls and reef islands developed on a submarine volcanic-ridge foundation in the Arabian Sea. The emergent islands are coral, unlike the tectonic island-arc setting of Andaman-Nicobar.

- [ANALYSIS] Geological origin shapes hazard: Andaman-Nicobar faces strong earthquake, tsunami and volcanic contexts; Lakshadweep's low coral islands face reef, freshwater-lens and sea-level vulnerability.

- [LIMIT] Detailed coral ecology and coastal regulation belong to Environment/Geography Topics 10-11. This package owns only tectonic foundation and origin distinction.

MUST-KNOW FACTS

1. [FACT] Andaman-Nicobar = active arc/forearc tectonic setting.

2. [FACT] Lakshadweep = coral atolls on submarine volcanic foundation.

3. [FACT] Indian coasts contain both rifted and sediment-loaded basin sectors.

UPSC TRAPS

WRONG: All Indian islands are coral.

CORRECT: Lakshadweep is coral; Andaman-Nicobar has an active tectonic arc setting.

WRONG: Andaman-Nicobar is a continental shield extension.

CORRECT: It belongs to an active plate-margin arc system.

MAINS ANGLE

[ANALYSIS] A two-column island comparison is a high-return Prelims and GS-I device.

STUDY LINK

Geography Topics 10-11; Environment Topic 24; Disaster Management tsunami owner.

HIGH

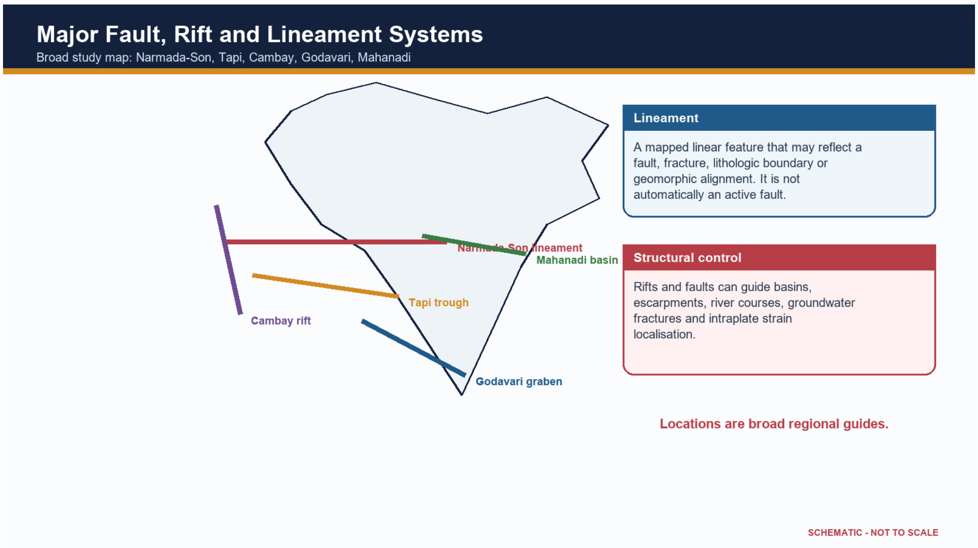
13. Major lineaments, faults and rifts

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] Peninsular India contains major inherited structural corridors including the Narmada-Son lineament, Tapi trough, Cambay rift and the Godavari and Mahanadi basin systems.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

System	Orientation/region	Key geological link
Narmada-Son	central India, broadly east-west	rift/lineament, basin, magmatic and seismic links
Tapi	west-central India	fault-controlled west-flowing trough
Cambay	Gujarat, broadly north-south	rift basin and hydrocarbon sedimentation
Godavari	south-central/east	Gondwana graben and coal-bearing basin
Mahanadi	east-central	Gondwana and coastal/offshore basin continuation

STATIC THEORY

- [FACT] The Narmada-Son structural zone crosses central India and is associated with faulting, rift-valley segments, basin development, magmatism and seismicity in different sectors.

- [FACT] Narmada and Tapi flow west through structurally controlled troughs. Their valleys cannot be explained by regional eastward slope alone.

- [FACT] The Cambay rift is a north-south basin in western India connected to sedimentation and hydrocarbon geology. Godavari and Mahanadi structural basins preserve Gondwana and younger sedimentary histories.

- [ANALYSIS] Lineaments can localise erosion, river courses, springs, groundwater fractures, basin margins and renewed stress. Some are composite zones with multiple episodes of reactivation.

- [LIMIT] A lineament mapped from imagery is not automatically a proven active fault. Activity requires field, geomorphic, geodetic and seismological evidence.

- [LIMIT] Schematic national maps must not imply exact fault traces or legal boundary precision.

MUST-KNOW FACTS

1. [FACT] Narmada and Tapi are structurally controlled west-flowing rivers.

2. [FACT] Rifts can host both sediments and later magmatism.

3. [FACT] Lineament does not automatically mean active fault.

UPSC TRAPS

WRONG: Every straight river segment proves active faulting.

CORRECT: Lithology, joints, inherited structures and slope can all create linear reaches.

WRONG: The Peninsula lacks major tectonic structures.

CORRECT: It contains old sutures, faults, rifts and lineaments.

MAINS ANGLE

[ANALYSIS] Use one map plus one mechanism paragraph; avoid asserting current activity without evidence.

STUDY LINK

Topic 05 drainage; Topic 03 intraplate seismicity; Topic 31 basin resources.

HIGH

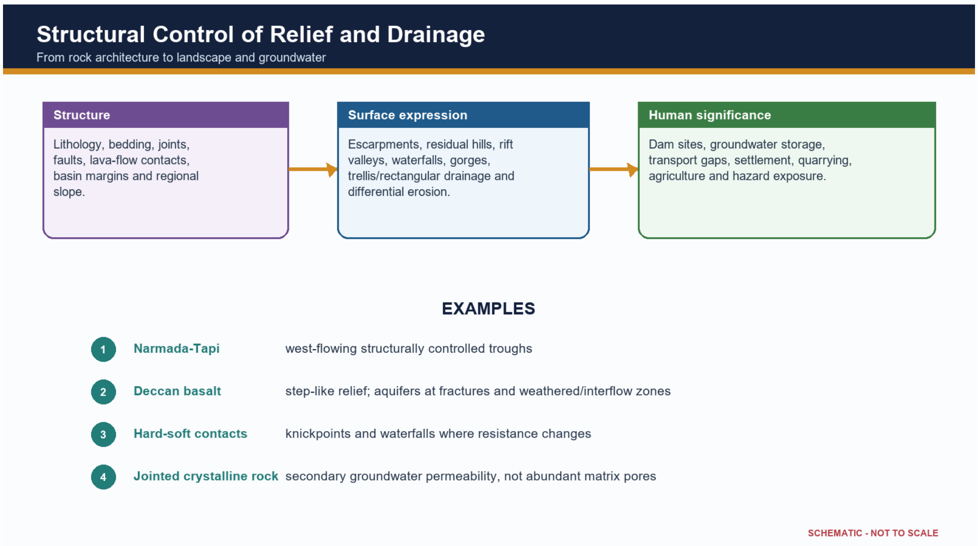
14. Structural control of relief, drainage, groundwater and waterfalls

GS-I / Prelims
Geography

NEWS TRIGGER

[ANALYSIS] Geological structure becomes examinable when it explains why relief, rivers, groundwater and resource patterns are spatially organised.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Structural element	Geomorphic/hydrologic expression	Named India cue
Rift/fault trough	linear valley and directed drainage	Narmada-Tapi
Joint/fracture	rectangular drainage and secondary aquifer	hard-rock Peninsula
Basalt flow contact	step relief and interflow aquifer	Deccan Plateau
Layer resistance contrast	cuesta/scarp/knickpoint	sedimentary plateau margins
Foreland alluvium	low gradient and high aquifer storage	Northern Plain

STATIC THEORY

- [FACT] Resistant quartzite, granite, gneiss or basalt can form uplands and scarps, while weaker shale or weathered zones erode into valleys. The result is differential erosion, not a simple hard-rock/soft-rock binary everywhere.

- [FACT] Faults and joints guide river segments and tributaries, producing rectangular or structurally aligned drainage. Alternating resistant and weak strata can support trellis patterns.

- [FACT] Waterfalls and knickpoints often occur at resistant rock bands, fault scarps, lava-flow contacts or rejuvenated reaches. Their persistence depends on erosion rates and downstream base level.

- [FACT] In hard-rock terrain groundwater is concentrated in weathered zones, fractures and contacts. In sedimentary basins, primary pore space can be more important; in alluvium, connected sand and gravel bodies form major aquifers.

- [ANALYSIS] Structural basins and lineaments influence dam siting, tunnelling, road stability and urban foundations. Engineering decisions require site investigation, not regional rock labels alone.

- [LIMIT] Detailed river courses, aquifer statistics and individual waterfall heights belong to later topics and current official datasets.

MUST-KNOW FACTS

1. [FACT] Structure guides but does not fully fix drainage.

2. [FACT] Hard-rock groundwater depends on secondary openings.

3. [FACT] Waterfalls can mark lithologic or structural breaks.

UPSC TRAPS

WRONG: Crystalline rock contains no groundwater.

CORRECT: Weathered and fractured zones can store and transmit water.

WRONG: Every waterfall lies on a fault.

CORRECT: Lithologic contrast, rejuvenation and lava contacts are alternative controls.

MAINS ANGLE

[ANALYSIS] Frame every example as structure -> process -> landform/water outcome -> local qualification.

STUDY LINK

Geography Topics 04-05; engineering geology is a supporting cross-link.

HIGH

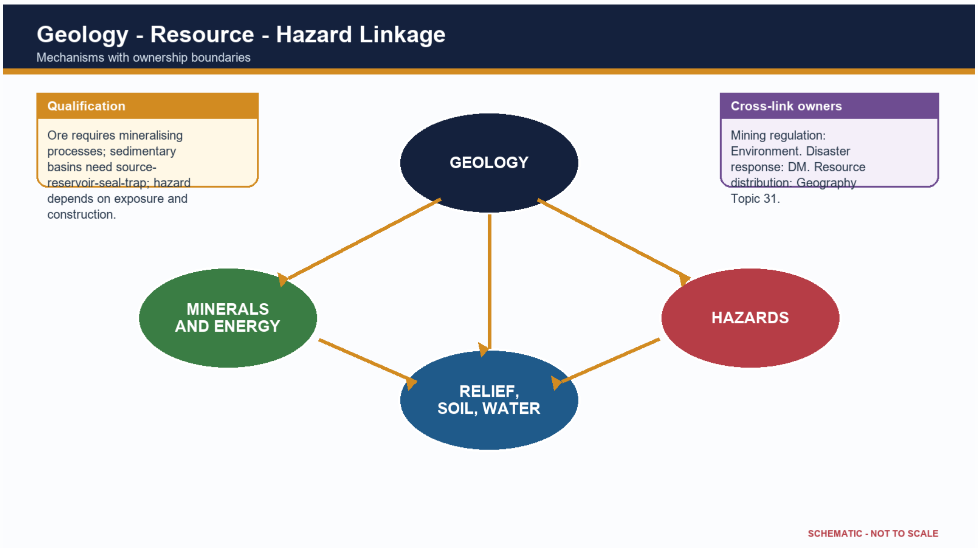
15. Geology, minerals, coal and petroleum: association without determinism

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] India's resource map reflects ancient shield mineralisation, Gondwana coal basins, sedimentary petroleum systems, coastal placers and weathering-derived ores.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Geological setting	Likely resource association	Essential qualification
Ancient shield/mobile belt	metallic ores	local mineralising event and grade
Gondwana rift basin	coal	coal-forming facies and preservation
Marine/deltaic basin	petroleum/gas	complete petroleum system
Lateritic surface	bauxite/iron enrichment	climate-drainage-parent control
Coastal placer	heavy minerals	erosion, transport and sorting

STATIC THEORY

- [FACT] Metallic ores such as iron, manganese, gold and base metals cluster in several ancient shield belts because of magmatic, hydrothermal, sedimentary and metamorphic histories.

- [FACT] Coal is concentrated in Gondwana structural basins; younger coal also occurs in parts of the north-east. Exact shares must be dated to official Ministry of Coal or GSI/IBM sources.

- [FACT] Petroleum and gas occur in sedimentary basins onshore and offshore where a complete petroleum system developed and survived.

- [FACT] Lateritic weathering can enrich bauxite or iron under suitable climate, drainage and parent material. Coastal sorting can concentrate heavy-mineral placers.

- [ANALYSIS] Geology creates potential; grade, depth, continuity, technology, infrastructure, environmental limits and prices determine whether a deposit becomes a resource or reserve.

- [LIMIT] Mining governance, EIA, rehabilitation and mine closure are Environment/Governance owners. Resource distribution and industrial linkage belong primarily to Topic 31.

- [FACT] The National Critical Mineral Mission was approved on 29 January 2025; it strengthens exploration, processing, recycling and supply security. It does not imply that all critical minerals lie in one geological province.

MUST-KNOW FACTS

1. [FACT] Ore deposit is not the same as economic reserve.
2. [FACT] Coal follows basin history; petroleum follows petroleum-system completeness.
3. [FACT] NCMM is a current policy anchor, not a geological classification.

UPSC TRAPS

WRONG: Old rocks are always mineral-rich.
CORRECT: Specific ore-forming and preservation processes are required.

WRONG: A sedimentary basin guarantees oil.
CORRECT: Source, maturation, reservoir, seal, trap and timing must align.

MAINS ANGLE

[ANALYSIS] Use the phrase 'geological necessary condition, not sufficient economic condition'.

STUDY LINK

Geography Topic 31; Environment EIA/mining; Economy critical-mineral value chains.

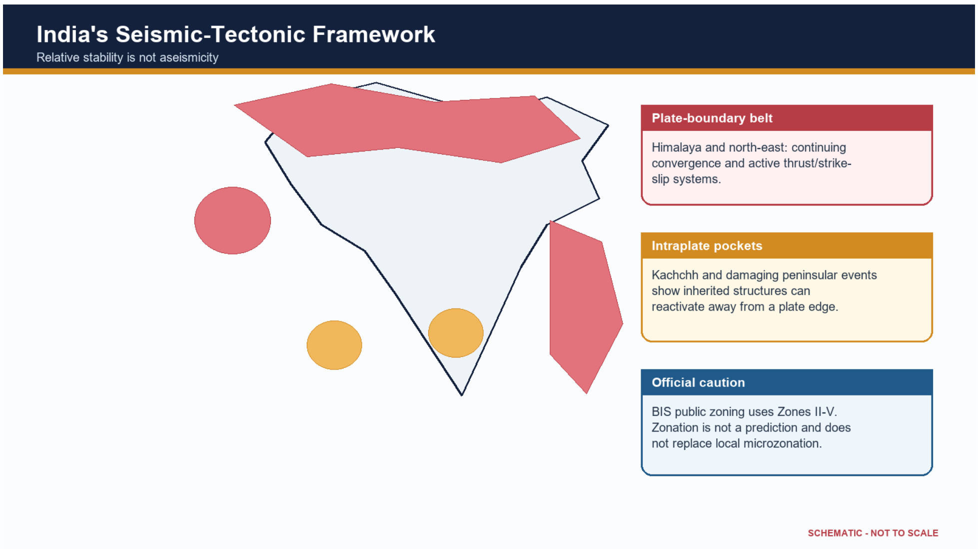
HIGH | 16. Seismicity: Himalaya, Kachchh and peninsular intraplate events

GS-I / Prelims
Geography

NEWS TRIGGER

[FACT] India's seismicity reflects active plate-boundary deformation in the Himalaya and north-east, subduction near Andaman-Nicobar, and intraplate reactivation in Kachchh and parts of the Peninsula.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Tectonic setting	India region	Principal geological control
Collision boundary	Himalaya/foothills	active thrusting and crustal shortening
Strike-slip/complex boundary	north-east	interaction of Himalayan and Indo-Burma systems
Subduction margin	Andaman-Nicobar	megathrust, arc and tsunami context
Intraplate rift/fault	Kachchh	reactivated inherited structures
Stable continental interior	Koyna-Latur-Jabalpur examples	local reactivation; event-specific mechanisms

STATIC THEORY

- [FACT] Himalayan earthquakes release strain on the Main Himalayan Thrust system and connected structures. Thick alluvium in adjoining plains can amplify shaking and liquefy under suitable saturation and grain conditions.

- [FACT] Kachchh is an intraplate high-hazard region with inherited rift faults and damaging historical earthquakes, including the 2001 Bhuj event.

- [FACT] Peninsular earthquakes such as Koyna, Latur and Jabalpur demonstrate that stable continental crust can reactivate along old weaknesses. Individual event causes require event-specific seismological evidence.

- [FACT] Andaman-Nicobar seismicity is linked to the Sunda-Andaman subduction system and includes tsunami-generating megathrust potential.

- [FACT] Current public BIS seismic zoning uses Zones II, III, IV and V. Zone V denotes the highest damage-risk category; there is no verified public Zone VI in the current map.

- [LIMIT] Zonation is a design/hazard generalisation, not an earthquake prediction. Local soil, basin geometry, building vulnerability and exposure require microzonation and site response analysis.

- [LIMIT] Disaster response doctrine, building-code enforcement and preparedness are owned by Disaster Management; geology provides the source and site-condition framework.

MUST-KNOW FACTS

1. [FACT] Relative stability is not aseismicity.
2. [FACT] Alluvium can amplify shaking.
3. [FACT] BIS public zoning remains II-V at the 2026 cutoff.

UPSC TRAPS

WRONG: Peninsular India cannot have damaging earthquakes.

CORRECT: Intraplate reactivation has produced destructive events.

WRONG: Seismic zone predicts the date and magnitude of the next quake.

CORRECT: It is a broad design/hazard category, not prediction.

MAINS ANGLE

[ANALYSIS] Explain source, path/site effect and vulnerability separately; do not equate tectonic activity with disaster magnitude.

STUDY LINK

Geography Topic 03; Disaster Management earthquake-resilient construction.

HIGH

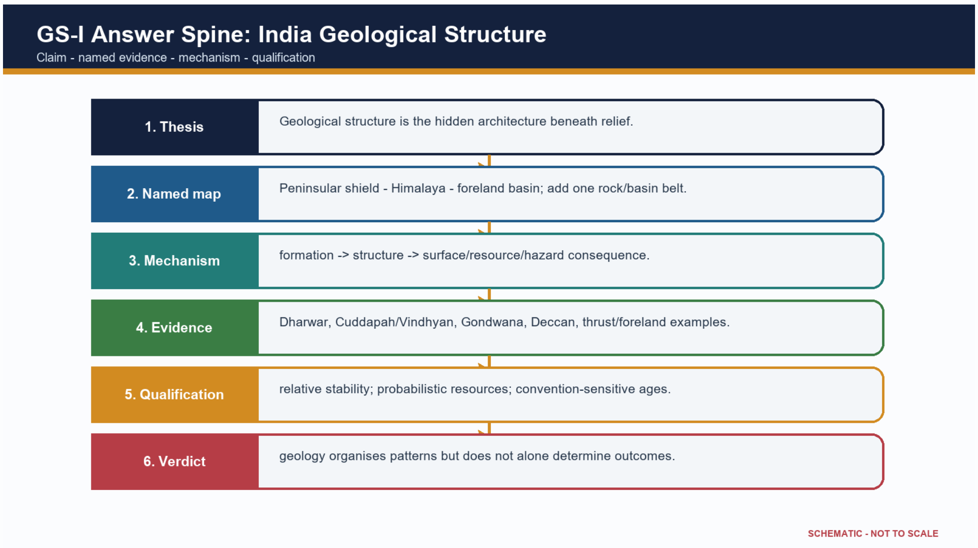
17. Regional comparisons, map skills and answer architecture

GS-I / Prelims
Geography

NEWS TRIGGER

[ANALYSIS] UPSC rewards a small accurate map, a causal chain and a qualification more than a long unsupported age table.

ORIGINAL STUDY VISUAL



Schematic/not to scale. Use as a conceptual map or cross-section, not legal cartography.

Original fact-checked study visual - schematic/not to scale.

KEY DATA

Demand	Best visual	Core evidence
Explain Himalayan structure	south-north cross-section	belts, thrusts, foreland basin
Relate geology to resources	rock-system-resource matrix	shield, Gondwana, basin examples
Assess stability	seismic-tectonic map	Himalaya, Kachchh, intraplate events
Compare island origins	two-column sketch	arc/subduction versus coral atoll
Discuss geological structure	three-province India map	Peninsula-Himalaya-plain causal chain

STATIC THEORY

- [ANALYSIS] Peninsula versus Himalaya: ancient composite shield with inherited structures versus active fold-thrust orogen; relative stability versus high convergence; exposed basement/mineral belts versus tectonic relief and sediment production.

- [ANALYSIS] Gondwana basin versus Deccan Trap: fault-bounded continental sedimentary basin with coal measures versus extrusive flood-basalt province with stacked flows.

- [ANALYSIS] Siwalik versus Indo-Gangetic alluvium: folded foreland molasse at the mountain front versus younger and continuing basin fill across the plain.

- [ANALYSIS] Andaman-Nicobar versus Lakshadweep: active subduction-related arc/forearc setting versus coral atolls on a submarine ridge foundation.

- [ANALYSIS] A 10-mark answer needs thesis, 4-5 evidence units and qualification; 15 marks needs map/cross-section and comparison; 20 marks needs regional variation, cross-links and a reasoned verdict.

- [LIMIT] A schematic map must be labelled not to scale and should avoid disputed political boundaries. Plot zones and structures, not false precision.

MUST-KNOW FACTS

1. [FACT] Map labels should support the argument.

2. [FACT] Cross-sections show mechanism better than decorative maps.

3. [FACT] Every model answer ends with a qualification.

UPSC TRAPS

WRONG: More labels always earn more marks.

CORRECT: Only relevant, correctly placed labels support the demand.

WRONG: A conclusion should repeat the introduction.

CORRECT: It should deliver a qualified verdict.

MAINS ANGLE

[ANALYSIS] Use the spine: claim -> named geological evidence/region -> mechanism/significance -> limitation/qualification -> verdict.

STUDY LINK

Final register notes below; Workbook map and cross-section drills.

HIGH

PYQ 2022 GS-I Q4 - Characteristics and types of primary rocks (10 marks)

GS-I / Prelims
Solved PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02.

STATIC THEORY

- Question/demand: Describe the characteristics and types of primary rocks.

- Model solution: [CLAIM] Primary rocks are igneous rocks formed by solidification of magma or lava and constitute the original mineral source for later sedimentary and metamorphic rocks. [EVIDENCE] Intrusive or plutonic rocks such as granite and gabbro cool slowly at depth and therefore commonly develop coarse crystals. [MECHANISM] Insulation delays cooling and permits crystal growth. [EVIDENCE] Extrusive rocks such as basalt and rhyolite cool rapidly at or near the surface and are commonly fine-grained; glassy or vesicular textures may occur when cooling is very rapid or gases escape. [MECHANISM] Composition ranges broadly from silica-rich felsic to iron-magnesium-rich mafic types, influencing colour, density and weathering. [EVIDENCE] Indian examples include the Deccan flood basalts and widespread granitic-gneissic terrains of the Peninsular shield. [LIMIT] 'Primary' describes genetic origin, not present exposure or economic value; later alteration can metamorphose an igneous protolith. [VERDICT] Thus texture, composition and mode of occurrence together classify primary rocks and explain their landscape and resource significance.

- Why this earns marks: it follows the directive 'describe', classifies both occurrence and composition, uses Indian evidence, gives mechanism and avoids confusing primary with oldest.

MUST-KNOW FACTS

1. Answer the exact directive and scope.

2. Use named geological evidence.

3. End with a qualification.

UPSC TRAPS

WRONG: Treat an adjacent PYQ as direct ownership.

CORRECT: Direct owner: Geography Core Topic 02.

MAINS ANGLE

Why this earns marks: it follows the directive 'describe', classifies both occurrence and composition, uses Indian evidence, gives mechanism and avoids confusing primary with oldest.

STUDY LINK

PYQ routing ledgers and local official paper/key text.

HIGH

PYQ 2025 GS-I Q16 - Continents and ocean basins due to crustal tectonics (15 marks)

GS-I / Prelims
Solved PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02; applied here to India's plate history.

STATIC THEORY

- Question/demand: Discuss the changes in the continents and ocean basins due to crustal tectonics.

- Model solution: [CLAIM] Continents and ocean basins are temporary configurations within a plate cycle rather than permanent containers. [EVIDENCE] Matching continental margins, continuous ancient rock belts, fossil distributions, palaeoclimate and palaeomagnetism establish former continental connections. [MECHANISM] At divergent boundaries, rifting and sea-floor spreading create new oceanic crust, progressing from continental rift to narrow sea and mature ocean. At convergent boundaries, subduction consumes oceanic lithosphere, forms trenches and arcs, and progressively closes basins. Continent-continent collision then shortens and thickens buoyant crust, producing suture zones and fold-thrust mountains. [EVIDENCE] The East African Rift, Red Sea, Atlantic, Pacific, Mediterranean and Himalayan-Tethyan suture illustrate successive states. India's separation from Gondwana, northward motion and collision with Eurasia demonstrate continental translation, Tethys closure, Himalayan uplift and foreland-basin creation. [ANALYSIS] Transform motion offsets ridges and margins without creating or destroying crust, while sedimentation reshapes passive margins and deltas. [LIMIT] The Wilson-cycle sequence is conceptual: rates vary, microplates and terranes complicate margins, and not every basin completes the full cycle. [VERDICT] Crustal tectonics continuously redistributes land and sea while organising global relief, resources and geophysical hazards.

- Why this earns marks: it answers both continents and oceans, integrates evidence with mechanism, uses India and world examples, and qualifies the idealised cycle.

MUST-KNOW FACTS

1. Answer the exact directive and scope.
2. Use named geological evidence.
3. End with a qualification.

UPSC TRAPS

WRONG: Treat an adjacent PYQ as direct ownership.

CORRECT: Direct owner: Geography Core Topic 02; applied here to India's plate history.

MAINS ANGLE

Why this earns marks: it answers both continents and oceans, integrates evidence with mechanism, uses India and world examples, and qualifies the idealised cycle.

STUDY LINK

PYQ routing ledgers and local official paper/key text.

HIGH

PYQ 2024 Prelims Q10 - Fold/block mountain matching

GS-I / Prelims
Solved PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. Official Set-A key: B.

STATIC THEORY

- Question/demand: Rows tested Vosges as fold, Alps as block, Appalachians as fold and Andes as fold; asked how many were correctly matched.

- Model solution: [FACT] Vosges is a block mountain, so row 1 is incorrect. Alps is a fold mountain, so row 2 is incorrect. Appalachians and Andes are fold-mountain systems, so rows 3 and 4 are correct. Therefore two rows are correct: option B.

- Elimination hinge: reverse the first two classifications; do not confuse a rift-flank horst with a young fold belt.

MUST-KNOW FACTS

1. Answer the exact directive and scope.

2. Use named geological evidence.

3. End with a qualification.

UPSC TRAPS

WRONG: Treat an adjacent PYQ as direct ownership.

CORRECT: Direct owner: Geography Core Topic 02. Official Set-A key: B.

MAINS ANGLE

Elimination hinge: reverse the first two classifications; do not confuse a rift-flank horst with a young fold belt.

STUDY LINK

PYQ routing ledgers and local official paper/key text.

HIGH | PYQ 2025 Prelims Q25 - Evidences of continental drift

GS-I / Prelims
Solved PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. Official Set-A key: C.

STATIC THEORY

- Question/demand: Statements covered matching ancient rocks of Brazil and western Africa, Ghana gold derived from the Brazil plateau when joined, and Gondwana sediment counterparts across southern landmasses.

- Model solution: [FACT] All three were accepted as evidence in the official key: geological continuity, placer/mineral-source reconstruction and cross-Gondwana stratigraphic correlation. Therefore option C.

- Elimination hinge: continental-drift evidence is broader than coastline fit; it includes lithologic, mineral, fossil, climatic and stratigraphic continuity.

MUST-KNOW FACTS

1. Answer the exact directive and scope.

2. Use named geological evidence.

3. End with a qualification.

UPSC TRAPS

WRONG: Treat an adjacent PYQ as direct ownership.

CORRECT: Direct owner: Geography Core Topic 02. Official Set-A key: C.

MAINS ANGLE

Elimination hinge: continental-drift evidence is broader than coastline fit; it includes lithologic, mineral, fossil, climatic and stratigraphic continuity.

STUDY LINK

PYQ routing ledgers and local official paper/key text.

HIGH | PYQ 2026 Prelims Q34 - Peninsular Block characteristics

GS-I / Prelims
Solved PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. PROVISIONAL ANSWER - NOT OFFICIALLY FINAL.

STATIC THEORY

- Question/demand: Statements: western-coast submergence due to tectonic activity; residual ranges such as Veliconda and Mahendragiri; deep V-shaped valleys formed by fast-flowing rivers.

- Model solution: [FACT] Statements 1 and 2 match standard descriptions of the Peninsular Block: parts of the west coast show submergence/faulted-margin expression, and residual hill ranges occur within the ancient denuded terrain. [ANALYSIS] Statement 3 is not a general Peninsular-block characteristic; mature, structurally controlled valleys are more typical than youthful deep V-shaped valleys as a defining trait. The provisional Set-A key records option B (1 and 2). [LIMIT] The 2026 key is provisional at the repository cutoff and must not be presented as a final UPSC key.

- Elimination hinge: distinguish isolated youthful incision from the broad geomorphic character of an old denuded shield.

MUST-KNOW FACTS

1. Answer the exact directive and scope.

2. Use named geological evidence.

3. End with a qualification.

UPSC TRAPS

WRONG: Treat an adjacent PYQ as direct ownership.

CORRECT: Direct owner: Geography Core Topic 02. PROVISIONAL ANSWER - NOT OFFICIALLY FINAL.

MAINS ANGLE

Elimination hinge: distinguish isolated youthful incision from the broad geomorphic character of an old denuded shield.

STUDY LINK

PYQ routing ledgers and local official paper/key text.

HIGH

Adjacent PYQ 2022 GS-I Q6 - Natural-resource potential of the Deccan Trap

GS-I / Prelims
Solved PYQ

NEWS TRIGGER

True owner: Geography Topic 31; included as a bounded geology application.

STATIC THEORY

- Question/demand: Discuss the natural resource potentials of the Deccan Trap.

- Model solution: [CLAIM] The Deccan Trap's resource potential arises from its layered basalt architecture and later weathering, not from basalt alone. [EVIDENCE] Weathered basalt contributes parent material to extensive black-soil tracts supporting cotton, oilseeds and other crops. [MECHANISM] Clay-rich weathering products retain moisture, though drainage and climate modify productivity. [EVIDENCE] Vesicular tops, fractures, weathered contacts and intertrappean beds form local aquifers; compact flow interiors may be poor aquifers. [EVIDENCE] Basalt is used as construction aggregate and building stone, while lateritic caps and associated weathering profiles can host local industrial-mineral potential. [ANALYSIS] Step relief, plateaux and escarpments create tourism and hydropower/transport opportunities in selected sectors. [LIMIT] Groundwater is heterogeneous, black soil can face waterlogging or cracking, and not all critical minerals are hosted by Deccan basalt. [VERDICT] The province is best understood as a soil-water-construction-landscape resource system whose benefits require location-specific management.

- Why this earns marks: it gives several resource dimensions, derives each from flow/weathering structure and avoids an exaggerated mineral list.

MUST-KNOW FACTS

1. Answer the exact directive and scope.

2. Use named geological evidence.

3. End with a qualification.

UPSC TRAPS

WRONG: Treat an adjacent PYQ as direct ownership.

CORRECT: True owner: Geography Topic 31; included as a bounded geology application.

MAINS ANGLE

Why this earns marks: it gives several resource dimensions, derives each from flow/weathering structure and avoids an exaggerated mineral list.

STUDY LINK

PYQ routing ledgers and local official paper/key text.

HIGH

Adjacent PYQ 2021 GS-I Q5 - Gondwanaland mineral base and mining share

GS-I / Prelims
Solved PYQ

NEWS TRIGGER

True owner: Geography Topic 31; geology application included without duplicating mining economics.

STATIC THEORY

- Question/demand: Discuss why India is well endowed with mineral resources due to Gondwanaland yet mining contributes relatively little to GDP.

- Model solution: [CLAIM] Gondwana-derived shield and basin history endowed India with diverse metallic ores and coal, but geological endowment is not identical to realised economic value. [EVIDENCE] Ancient Peninsular mobile belts host iron, manganese and base-metal provinces, while faulted Gondwana basins host major coalfields. [MECHANISM] Magmatism, metamorphism, hydrothermal activity, sedimentation and long preservation created deposits. [ANALYSIS] Low value addition, variable ore grade, deep or fragmented deposits, infrastructure gaps, environmental-social constraints, exploration uncertainty and cyclical prices can restrict mining's GDP contribution. [LIMIT] Gondwanaland should not be used as a blanket explanation for every mineral or petroleum basin, and current GDP shares require dated official data. [VERDICT] Policy must convert geological knowledge into responsible exploration, beneficiation, recycling and community-sensitive value chains rather than equating extraction volume with development.

- Why this earns marks: it separates endowment from economic conversion, uses geological mechanisms and preserves the true Topic 31 ownership boundary.

MUST-KNOW FACTS

1. Answer the exact directive and scope.

2. Use named geological evidence.

3. End with a qualification.

UPSC TRAPS

WRONG: Treat an adjacent PYQ as direct ownership.

CORRECT: True owner: Geography Topic 31; geology application included without duplicating mining economics.

MAINS ANGLE

Why this earns marks: it separates endowment from economic conversion, uses geological mechanisms and preserves the true Topic 31 ownership boundary.

STUDY LINK

PYQ routing ledgers and local official paper/key text.

HIGH

Learning MCQ Loop 1-4

Prelims Concept practice

NEWS TRIGGER

Correct keys follow strict A-B-C-D rotation.

HIGH

PYQ 2018 Prelims Q57 - Magnetic reversal and early atmosphere

Prelims Solved verified-demand PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY.

STATIC THEORY

- Question/demand: Statements tested whether Earth's magnetic field has reversed over geological time, whether the earliest atmosphere had 54 per cent oxygen and no carbon dioxide, and whether living organisms modified the early atmosphere.

- [FACT] Statement 1 is correct in concept: palaeomagnetic records preserve repeated polarity reversals, although reversal intervals are irregular and 'every few hundred thousand years' is only a broad formulation.

- [FACT] Statement 2 is false: the early Earth did not possess an oxygen-rich atmosphere of the stated composition; substantial free oxygen accumulated much later.

- [FACT] Statement 3 is correct: photosynthetic life profoundly altered atmospheric chemistry by producing oxygen and participating in long-term carbon cycling.

- INFERRED ANSWER: statements 1 and 3 only, option C. The local repository does not hold a readable official 2018 key.

MUST-KNOW FACTS

1. Magnetic reversals are recorded by remanent magnetism in suitable rocks.

2. The early atmosphere was not 54 per cent oxygen.

3. Life, especially photosynthesis, modified atmospheric composition.

UPSC TRAPS

WRONG: Present option C as locally official.

CORRECT: Label it inferred because the local official key is unavailable.

WRONG: Treat reversal spacing as perfectly periodic.

CORRECT: Reversal intervals are irregular.

MAINS ANGLE

Use palaeomagnetism as evidence for geologic change and plate reconstruction.

STUDY LINK

Core Topic 02 evidence and dating.

HIGH

PYQ 2023 Prelims Q9 - Amarkantak, Biligirirangan and Seshachalam

Prelims Solved locally held PYQ

NEWS TRIGGER

Direct owner: Geography Core Topic 02. INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY.

STATIC THEORY

- Question/demand: Statements claimed Amarkantak lies at the confluence of Vindhya and Sahyadri, Biligirirangan is the easternmost Satpura, and Seshachalam is the southernmost Western Ghats.

- [FACT] Statement 1 is incorrect: Amarkantak is associated with the Vindhya-Satpura meeting region, not the Sahyadri/Western Ghats.

- [FACT] Statement 2 is incorrect: Biligirirangan lies in southern Karnataka at an Eastern Ghats-Western Ghats biogeographic linkage, not the eastern Satpura.

- [FACT] Statement 3 is incorrect: Seshachalam is part of the Eastern Ghats in Andhra Pradesh, not the southern Western Ghats.

- INFERRED ANSWER: none of the statements is correct, option D. The local question text is held, but the routing ledger records the official key as unavailable.

MUST-KNOW FACTS

1. Amarkantak: Vindhya-Satpura highland context.

2. Biligirirangan: southern Karnataka, Eastern-Western Ghats linkage.

3. Seshachalam: Eastern Ghats, Andhra Pradesh.

UPSC TRAPS

WRONG: Attach a true hill name to a plausible but wrong parent range.

CORRECT: Locate each hill independently on a mental map.

WRONG: Present option D as locally official.

CORRECT: Keep the inferred-key label.

MAINS ANGLE

Map-based elimination tests geological and physiographic regional placement.

STUDY LINK

Peninsular shield map and regional-comparison practice.

STATIC THEORY

- MCQ 1: Which statement best distinguishes geological structure from physiography? Options: A. Structure concerns rock architecture; physiography concerns broad surface divisions | B. Both terms mean relief only | C. Physiography concerns mineral chemistry only | D. Structure concerns climate zones

- MCQ 2: Which is the safest description of the Peninsular block? Options: A. Uniform Archaean slab | B. Composite ancient shield cut by later rifts and partly covered by younger rocks | C. Young volcanic arc | D. Sediment-filled foredeep

- MCQ 3: Which evidence directly helps reconstruct past plate latitude? Options: A. Modern rainfall | B. River discharge | C. Palaeomagnetism | D. Population density

- MCQ 4: Which statement about radiometric dating is correct? Options: A. It always dates sediment deposition | B. It makes stratigraphy unnecessary | C. It gives the same age for all minerals | D. It dates a mineral event that needs geological interpretation

- SOLUTIONS

- 1-A: Structure is the hidden rock/tectonic framework; physiography is its broad surface expression.

- 2-B: The Peninsula is a mosaic of cratons/mobile belts with later basins, faults and lava cover.

- 3-C: Magnetic remanence in rocks records ancient field orientation after careful correction.

- 4-D: The measured isotopic clock may represent crystallisation, metamorphism or cooling.

MUST-KNOW FACTS

1. Attempt before reading the solution.

2. Explain why each distractor fails.

UPSC TRAPS

WRONG: Memorise the option letter only.

CORRECT: Retain the geological distinction used for elimination.

MAINS ANGLE

Convert each MCQ distinction into one map label or Mains evidence unit.

STUDY LINK

Workbook contains expanded and remedial practice.

HIGH

Learning MCQ Loop 5-8

Prelims Concept practice

NEWS TRIGGER

Correct keys follow strict A-B-C-D rotation.

STATIC THEORY

- MCQ 5: Gondwana in 'Gondwana Supergroup' refers primarily to Options: A. a continental sedimentary succession in faulted basins | B. the present Indian plate boundary | C. the Deccan lava province | D. the Himalayan crystalline core

- MCQ 6: The main resource linkage of classic Gondwana basins is Options: A. offshore petroleum only | B. major coal measures in continental rift basins | C. coral limestone | D. gold-bearing greenstone belts

- MCQ 7: Which pair is correctly matched? Options: A. Vindhyan-flood basalt | B. Dharwar-modern alluvium | C. Deccan Trap-stacked basalt flows | D. Siwalik-oldest craton

- MCQ 8: Which is NOT sufficient to infer petroleum? Options: A. sedimentary basin | B. source rock | C. reservoir and seal | D. basin presence alone

- SOLUTIONS

- 5-A: The stratigraphic succession is distinct from Gondwana the palaeocontinent.

- 6-B: Coal-bearing sediment cycles accumulated in subsiding continental basins.

- 7-C: Deccan Trap is an extrusive flood-basalt province.

- 8-D: A complete petroleum system and timing are required.

MUST-KNOW FACTS

1. Attempt before reading the solution.

2. Explain why each distractor fails.

UPSC TRAPS

WRONG: Memorise the option letter only.

CORRECT: Retain the geological distinction used for elimination.

MAINS ANGLE

Convert each MCQ distinction into one map label or Mains evidence unit.

STUDY LINK

Workbook contains expanded and remedial practice.

HIGH

Learning MCQ Loop 9-12

Prelims Concept practice

NEWS TRIGGER

Correct keys follow strict A-B-C-D rotation.

STATIC THEORY

- MCQ 9: Siwalik rocks are best described as Options: A. young foreland molasse folded at the Himalayan front | B. oldest Himalayan crystalline basement | C. oceanic crust of the Tethys | D. Deccan basalt outliers

- MCQ 10: Which order is broadly south-to-north? Options: A. Greater-Lesser-Siwalik-plain | B. Siwalik-Lesser-Greater-Tethyan/Trans-Himalaya | C. Tethys-Siwalik-plain-Greater | D. Plain-Trans-Himalaya-Siwalik-Lesser

- MCQ 11: The Indo-Ganga-Brahmaputra Plain is structurally Options: A. an exposed craton | B. a volcanic plateau | C. a sediment-filled foreland basin | D. an island arc

- MCQ 12: Which distinction is correct? Options: A. Both island groups are coral | B. Lakshadweep is an active arc | C. Andaman is a shield fragment | D. Andaman-Nicobar is arc-related; Lakshadweep is coral-atoll terrain

- SOLUTIONS

- 9-A: They are erosion products of the rising Himalaya deposited in a foreland setting.

- 10-B: The simplified cross-section runs outer foothills to higher crystalline and northern domains.

- 11-C: Flexure under Himalayan loading created accommodation filled by alluvium.

- 12-D: Their geological settings and hazards differ fundamentally.

MUST-KNOW FACTS

1. Attempt before reading the solution.

2. Explain why each distractor fails.

UPSC TRAPS

WRONG: Memorise the option letter only.

CORRECT: Retain the geological distinction used for elimination.

MAINS ANGLE

Convert each MCQ distinction into one map label or Mains evidence unit.

STUDY LINK

Workbook contains expanded and remedial practice.

HIGH

Learning MCQ Loop 13-16

Prelims Concept practice

NEWS TRIGGER

Correct keys follow strict A-B-C-D rotation.

STATIC THEORY

- MCQ 13: The Narmada and Tapi are high-yield examples of Options: A. structurally controlled west-flowing trough rivers | B. glacial antecedent rivers | C. delta-building east-flowing rivers | D. coral-island streams

- MCQ 14: Groundwater in crystalline hard rock commonly depends on Options: A. abundant primary pore space | B. weathering, joints and fractures | C. coal seams only | D. marine fossils

- MCQ 15: Which statement best controls resource determinism? Options: A. Every old rock is an ore | B. Every basin has petroleum | C. Geology creates potential but grade, system completeness and economics decide resources | D. Minerals follow state boundaries

- MCQ 16: Which is the correct seismic caution? Options: A. Peninsula is aseismic | B. Zone V predicts the next earthquake | C. Alluvium always protects buildings | D. Relative stability does not exclude damaging intraplate earthquakes

- SOLUTIONS

- 13-A: They occupy major structural troughs across the regional slope.

- 14-B: Secondary permeability controls most hard-rock aquifers.

- 15-C: Geological association is necessary in many cases but not economically sufficient.

- 16-D: Kachchh, Latur, Koyna and Jabalpur show inherited structures can reactivate.

MUST-KNOW FACTS

1. Attempt before reading the solution.

2. Explain why each distractor fails.

UPSC TRAPS

WRONG: Memorise the option letter only.

CORRECT: Retain the geological distinction used for elimination.

MAINS ANGLE

Convert each MCQ distinction into one map label or Mains evidence unit.

STUDY LINK

Workbook contains expanded and remedial practice.

HIGH

10 marks: Distinguish India's geological structure from its physiographic divisions. Illustrate with two regions.

GS-I Solved Mains practice

NEWS TRIGGER

Model follows claim -> named evidence/region -> mechanism/significance -> limitation/qualification.

STATIC THEORY

- [CLAIM] Geological structure is the rock-and-tectonic architecture beneath India, whereas physiography is its broad surface expression. [EVIDENCE] The Peninsular Plateau is a physiographic unit, but geologically it contains Dharwar, Bastar, Singhbhum and Bundelkhand blocks, Proterozoic basins, Gondwana rifts and Deccan basalt. [MECHANISM] Differential resistance, faulting and long denudation generate residual hills, scarps and structurally guided rivers. [EVIDENCE] The Northern Plain appears as level physiography but geologically is a foreland depression filled by Himalayan and peninsular alluvium. [MECHANISM] Orogenic loading caused flexure and sediment created the plain. [LIMIT] Climate and rivers continue to reshape both provinces; structure is a template, not the sole cause. [VERDICT] Thus physiography is best read as the visible, modified expression of deeper geology. Why this earns marks: it defines both terms, uses two named regions, explains mechanisms and qualifies determinism.

MUST-KNOW FACTS

1. Scale evidence to marks.

2. Use a labelled schematic where relevant.

3. End with 'Why this earns marks'.

UPSC TRAPS

WRONG: List systems without mechanism.

CORRECT: Connect each named region to structure and consequence.

MAINS ANGLE

it defines both terms, uses two named regions, explains mechanisms and qualifies determinism.

STUDY LINK

Use Visual 16 as the answer spine.

HIGH

15 marks: Explain how the geological evolution of Peninsular India controls coal, metallic minerals and groundwater.

GS-I Solved Mains practice

NEWS TRIGGER

Model follows claim -> named evidence/region -> mechanism/significance -> limitation/qualification.

STATIC THEORY

- [CLAIM] Peninsular resource geography reflects distinct stages of crustal evolution rather than one uniformly mineral-rich shield. [EVIDENCE] Ancient Dharwar-Singhbhum-Aravalli mobile belts experienced magmatism, metamorphism and hydrothermal activity, creating iron, manganese, gold and base-metal provinces. [MECHANISM] Ore localisation followed specific mineralising events and structural traps. [EVIDENCE] Later faulting formed Damodar, Son-Mahanadi and Godavari Gondwana basins where continental sediments and organic matter accumulated. [MECHANISM] Subsidence, burial and preservation produced major coal measures. [EVIDENCE] Crystalline rocks have low primary porosity, but weathered mantles, joints and faults store groundwater; Deccan basalt adds vesicular tops, fractures and interflow contacts. [ANALYSIS] Therefore metallic ore, coal and aquifer maps differ because their geological mechanisms differ. [LIMIT] Grade, recharge, technology, environmental constraints and economics decide usable resources. [VERDICT] The Peninsula is best understood as a layered resource system: ancient mineralised crust, younger sedimentary basins and structurally controlled aquifers. Why this earns marks: it covers all three demands with named belts, distinct mechanisms and an economic-hydrological qualification.

MUST-KNOW FACTS

1. Scale evidence to marks.

2. Use a labelled schematic where relevant.

3. End with 'Why this earns marks'.

UPSC TRAPS

WRONG: List systems without mechanism.

CORRECT: Connect each named region to structure and consequence.

MAINS ANGLE

it covers all three demands with named belts, distinct mechanisms and an economic-hydrological qualification.

STUDY LINK

Use Visual 16 as the answer spine.

HIGH

20 marks: 'India's geological stability is relative, spatially uneven and socially consequential.' Analyse.

GS-I Solved Mains practice

NEWS TRIGGER

Model follows claim -> named evidence/region -> mechanism/significance -> limitation/qualification.

STATIC THEORY

- [CLAIM] India's geological stability is a comparison of deformation regimes, not a division between active and inactive territory. [EVIDENCE] The Himalaya and north-east occupy an active collision system with thrusting, crustal shortening and frequent earthquakes. [MECHANISM] Continuing India-Eurasia convergence stores and releases elastic strain; adjoining alluvial basins can amplify shaking. [EVIDENCE] Andaman-Nicobar lies on a subduction-related arc with earthquake, tsunami and volcanic contexts. [EVIDENCE] Peninsular cratons are older and less deforming on average, yet inherited rifts and faults produce intraplate hazards in Kachchh and events such as Koyna, Latur and Jabalpur. [MECHANISM] Far-field stress and local structural weaknesses can reactivate old zones. [ANALYSIS] Social consequence depends on exposure: dense Himalayan towns, vulnerable masonry, unstable slopes, dams, industrial corridors and soft-sediment cities transform geological processes into risk. Resource consequences are also uneven: shield belts host metallic ores, Gondwana rifts coal, and sedimentary margins petroleum systems. [LIMIT] Seismic zonation does not predict earthquakes, and resource association does not ensure economic extraction. [VERDICT] Planning must replace the myth of a stable versus unstable India with region-specific microzonation, engineering investigation, responsible resource assessment and ecosystem-sensitive land use. Why this earns marks: it interprets 'relative', compares four tectonic settings, integrates resource and risk consequences, and ends with a qualified policy verdict.

MUST-KNOW FACTS

1. Scale evidence to marks.

2. Use a labelled schematic where relevant.

3. End with 'Why this earns marks'.

UPSC TRAPS

WRONG: List systems without mechanism.

CORRECT: Connect each named region to structure and consequence.

MAINS ANGLE

it interprets 'relative', compares four tectonic settings, integrates resource and risk consequences, and ends with a qualified policy verdict.

STUDY LINK

Use Visual 16 as the answer spine.

HIGH

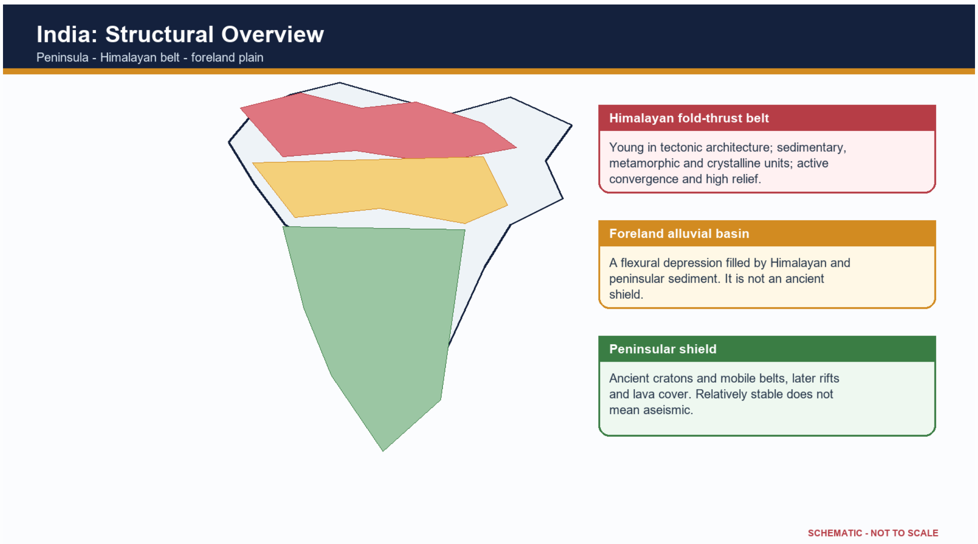
Final Consolidated Register Notes 1/4 - National skeleton and chronology

GS-I / Prelims
Final register notes

NATIONAL SKELETON AND CHRONOLOGY - REVISION FOCUS

Compressed topic-specific revision after all teaching and practice.

VISUAL SPINE



Schematic/not to scale.

Original study visual.

NATIONAL SKELETON AND CHRONOLOGY - CAUSAL AND COMPARATIVE LOGIC

- [FACT/ANALYSIS] Three provinces: composite Peninsular shield; active Himalayan fold-thrust belt; sediment-filled Indo-Ganga-Brahmaputra foreland basin.

- [FACT/ANALYSIS] Broad order: Archaean-Dharwar -> Cuddapah/Vindhyan -> Gondwana -> Deccan volcanism -> Himalayan collision/Cenozoic basins -> Quaternary alluvium.

- [FACT/ANALYSIS] Terminology and ages vary; state the scheme and prioritise relative order.

- [FACT/ANALYSIS] Stable Peninsula is relative, not aseismic or structurally uniform.

NATIONAL SKELETON AND CHRONOLOGY - RAPID RECALL

1. Three provinces: composite Peninsular shield; active Himalayan fold-thrust belt; sediment-filled Indo-Ganga-Brahmaputra foreland basin.

2. Broad order: Archaean-Dharwar -> Cuddapah/Vindhyan -> Gondwana -> Deccan volcanism -> Himalayan collision/Cenozoic basins -> Quaternary alluvium.

3. Terminology and ages vary; state the scheme and prioritise relative order.

4. Stable Peninsula is relative, not aseismic or structurally uniform.

NATIONAL SKELETON AND CHRONOLOGY - ERROR CHECKS

WRONG: Use an absolute or deterministic formulation.

CORRECT: State the mechanism and its geological qualification.

NATIONAL SKELETON AND CHRONOLOGY - PYQ AND ANSWER ROUTE

Use as the final 3-minute answer plan.

NATIONAL SKELETON AND CHRONOLOGY - NEXT REVISION LINK

Revisit full section when a causal link is unclear.

HIGH

Final Consolidated Register Notes 2/4 - Rock systems and basin logic

GS-I / Prelims
Final register notes

ROCK SYSTEMS AND BASIN LOGIC - REVISION FOCUS

Compressed topic-specific revision after all teaching and practice.

VISUAL SPINE

Rock-System - Age - Resource Matrix

Associations are probabilistic, not deterministic

System / setting	Broad age-status	Typical rocks	Bounded resource linkage
Archaean-Dharwar shield	Precambrian; schemes vary	gneiss, granite, schist, greenstone	metallic-mineral provinces; not every belt mineralised
Cuddapah-Vindhyan	Proterozoic / Purana	sandstone, shale, limestone, quartzite	stone, limestone; Panna diamond association
Gondwana basins	late Palaeozoic-Mesozoic succession	continental sandstone, shale, coal measures	major coalfields in faulted basins
Deccan Trap	late Cretaceous-earliest Cenozoic transition	stacked basalt flows	soil, groundwater and construction links; local mineralisation
Cenozoic basins	Tertiary-Quaternary	marine/continental sediment and alluvium	petroleum where source-reservoir-seal-trap coexist

SCHEMATIC - NOT TO SCALE

Schematic/not to scale.

Original study visual.

ROCK SYSTEMS AND BASIN LOGIC - CAUSAL AND COMPARATIVE LOGIC

- [FACT/ANALYSIS] Dharwar is an ancient supracrustal/metamorphic belt term, not every Archaean rock.

- [FACT/ANALYSIS] Cuddapah and Vindhyan are Proterozoic sedimentary basins over older basement.

- [FACT/ANALYSIS] Gondwana is a faulted continental sedimentary succession with major coal measures.

- [FACT/ANALYSIS] Deccan Trap is stacked flood basalt; black-soil and aquifer links require pedogenic/structural qualification.

- [FACT/ANALYSIS] Petroleum needs source-reservoir-seal-trap-timing; basin presence alone is insufficient.

ROCK SYSTEMS AND BASIN LOGIC - RAPID RECALL

1. Dharwar is an ancient supracrustal/metamorphic belt term, not every Archaean rock.

2. Cuddapah and Vindhyan are Proterozoic sedimentary basins over older basement.

3. Gondwana is a faulted continental sedimentary succession with major coal measures.

4. Deccan Trap is stacked flood basalt; black-soil and aquifer links require pedogenic/structural qualification.

5. Petroleum needs source-reservoir-seal-trap-timing; basin presence alone is insufficient.

ROCK SYSTEMS AND BASIN LOGIC - ERROR CHECKS

WRONG: Use an absolute or deterministic formulation.

CORRECT: State the mechanism and its geological qualification.

ROCK SYSTEMS AND BASIN LOGIC - PYQ AND ANSWER ROUTE

Use as the final 3-minute answer plan.

ROCK SYSTEMS AND BASIN LOGIC - NEXT REVISION LINK

Revisit full section when a causal link is unclear.

HIGH

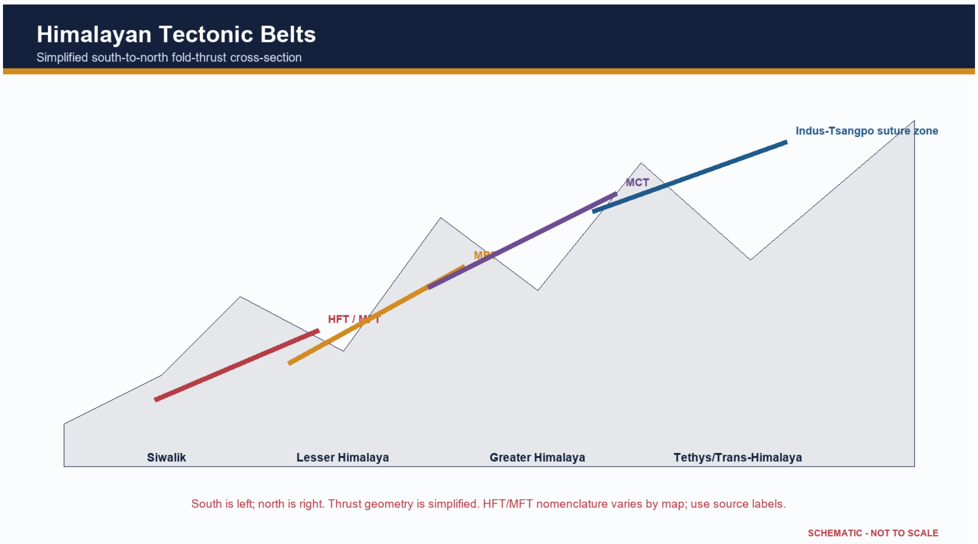
Final Consolidated Register Notes 3/4 - Himalaya, islands and seismicity

GS-I / Prelims
Final register notes

HIMALAYA, ISLANDS AND SEISMICITY - REVISION FOCUS

Compressed topic-specific revision after all teaching and practice.

VISUAL SPINE



Schematic/not to scale.

Original study visual.

HIMALAYA, ISLANDS AND SEISMICITY - CAUSAL AND COMPARATIVE LOGIC

- [FACT/ANALYSIS] Simplified south-north: foreland -> Siwalik -> Lesser -> Greater -> Tethyan/Trans-Himalayan and suture domains.

- [FACT/ANALYSIS] Siwalik is young foreland molasse, not oldest Himalaya; the orogen is not one simple fold.

- [FACT/ANALYSIS] Andaman-Nicobar is arc/forearc; Lakshadweep is coral-atoll terrain on submarine volcanic foundation.

- [FACT/ANALYSIS] Himalaya, north-east, Andaman and Kachchh are major tectonic contexts; Peninsula still has damaging intraplate events.

- [FACT/ANALYSIS] BIS public map uses Zones II-V; zonation is not prediction.

HIMALAYA, ISLANDS AND SEISMICITY - RAPID RECALL

1. Simplified south-north: foreland -> Siwalik -> Lesser -> Greater -> Tethyan/Trans-Himalayan and suture domains.

2. Siwalik is young foreland molasse, not oldest Himalaya; the orogen is not one simple fold.

3. Andaman-Nicobar is arc/forearc; Lakshadweep is coral-atoll terrain on submarine volcanic foundation.

4. Himalaya, north-east, Andaman and Kachchh are major tectonic contexts; Peninsula still has damaging intraplate events.

5. BIS public map uses Zones II-V; zonation is not prediction.

HIMALAYA, ISLANDS AND SEISMICITY - ERROR CHECKS

WRONG: Use an absolute or deterministic formulation.

CORRECT: State the mechanism and its geological qualification.

HIMALAYA, ISLANDS AND SEISMICITY - PYQ AND ANSWER ROUTE

Use as the final 3-minute answer plan.

HIMALAYA, ISLANDS AND SEISMICITY - NEXT REVISION LINK

Revisit full section when a causal link is unclear.

HIGH

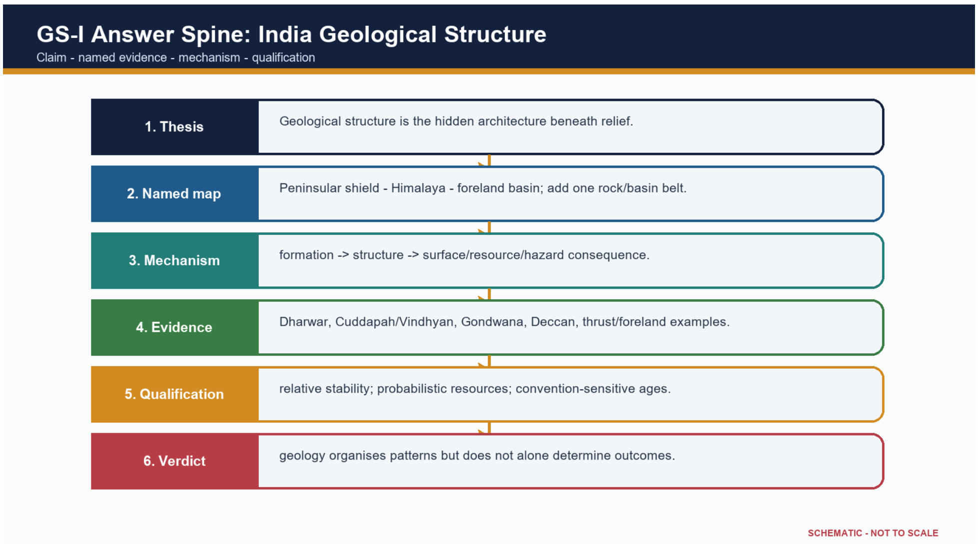
Final Consolidated Register Notes 4/4 - Causal chains, traps and answer spine

GS-I / Prelims
Final register notes

CAUSAL CHAINS, TRAPS AND ANSWER SPINE - REVISION FOCUS

Compressed topic-specific revision after all teaching and practice.

VISUAL SPINE



Schematic/not to scale.
Original study visual.

CAUSAL CHAINS, TRAPS AND ANSWER SPINE - CAUSAL AND COMPARATIVE LOGIC

- [FACT/ANALYSIS] Structure -> differential erosion/drainage/groundwater -> resource and hazard pattern.

- [FACT/ANALYSIS] Geology creates potential; grade, preservation, technology, exposure and governance mediate outcomes.

- [FACT/ANALYSIS] Answer spine: claim -> named evidence -> mechanism/significance -> limitation -> verdict.

- [FACT/ANALYSIS] Direct PYQs: 2022 primary rocks; 2024 mountains; 2025 drift and tectonics; 2026 Peninsular Block provisional.

- [FACT/ANALYSIS] Error checks: Deccan volcanic; Gondwana coal rift-linked; plain is foreland fill; Peninsula not aseismic; island origins differ.

CAUSAL CHAINS, TRAPS AND ANSWER SPINE - RAPID RECALL

1. Structure -> differential erosion/drainage/groundwater -> resource and hazard pattern.

2. Geology creates potential; grade, preservation, technology, exposure and governance mediate outcomes.

3. Answer spine: claim -> named evidence -> mechanism/significance -> limitation -> verdict.

4. Direct PYQs: 2022 primary rocks; 2024 mountains; 2025 drift and tectonics; 2026 Peninsular Block provisional.

5. Error checks: Deccan volcanic; Gondwana coal rift-linked; plain is foreland fill; Peninsula not aseismic; island origins differ.

CAUSAL CHAINS, TRAPS AND ANSWER SPINE - ERROR CHECKS

WRONG: Use an absolute or deterministic formulation.

CORRECT: State the mechanism and its geological qualification.

End of Register Notes - UPSC Agent / Copilot CLI